



Modeling the impact of harmful algal blooms on aquatic life and human health

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Abstract Harmful algal blooms pose a significant and growing threat to aquatic ecosystems worldwide, largely driven by excessive nutrient enrichment. These blooms disrupt ecological balance, degrade water quality, and have profound socioeconomic consequences. To address this pressing issue, this study presents a mathematical model that captures the complex interplay between nutrients, algae, fish, and human population. The model incorporates critical ecological processes, including algae growth dynamics, nutrient cycling, predation of algae by fish, and the detrimental effects of harmful algal blooms on fish and human population. By integrating these interactions, the model explores the feedback mechanisms that govern the ecosystem's stability and resilience. Using bifurcation analysis, the model identifies key thresholds and critical points that determine shifts in the system's dynamical behavior, revealing the presence of saddle-node, transcritical, and Hopf bifurcations. These findings highlight the delicate balance within aquatic ecosystems and the potential for abrupt transitions between stable states under varying environmental conditions. This study provides a robust theoretical foundation for understanding the mechanisms driving harmful algal blooms' formation and persistence. It emphasizes the need for integrated management strategies that address both ecological and human dimensions, offering insights into sustainable approaches for mitigating the impacts of harmful algal blooms.

1 Introduction

The gradual degradation of Earth's freshwater systems due to excessive algae proliferation poses a significant environmental challenge [1, 2]. This degradation stems from the over-saturation of water bodies with nutrients, particularly nitrogen and phosphorus, which creates congruent conditions for algae proliferation [3, 4]. Anthropogenic activities, specifically agricultural run-off, urban waste water discharge, and industrial effluents, are major contributors to this nutrient enrichment. As these nutrients accumulate, they trigger eutrophication, leading to proliferation of harmful algal blooms (HABs). HABs encompass a diverse group of algae species that, when proliferate, induce severe ecological and socioeconomic impacts, including human health risks due to consumption of contaminated seafood and water, exposure to toxic aerosols, large-scale fish mortality, and ecosystem's degradation due to excessive biomass accumulation [5–7]. Additionally, exposure to toxic aerosols generated by HABs can cause respiratory issues and skin irritations in humans [8]. These harmful effects indicate that if HABs proliferation is left unaddressed, the continued decline in freshwater quality will jeopardize ecosystem's stability and pose significant risks to both environment and human well-beings.

In response to the growing challenge of algal bloom, the scientific community has initiated extensive research efforts to elucidate the complex dynamics underlying these blooms and their far-reaching ecological and socioeconomic impacts [9–20]. In the studies of Truscott and Brindley [21, 22], a theoretical approach has been used to interpret bloom development with the application of the excitable media theory. This framework has been adapted to explain how aquatic ecosystems respond to external nutrient inputs and environmental perturbations. Moreover, building on this theoretical foundation, Truscott [23] further expanded the application of excitable media theory to model the sudden emergence of phytoplankton peaks. His study demonstrated that when microalgal growth rates exceed a critical threshold, an abrupt increase in phytoplankton density occurs. This phenomenon suggests that minor changes in environmental conditions, such as nutrient influxes, temperature shifts, or hydrodynamic variations, can trigger rapid algae proliferation. Recently, Maurya et al. [16] investigated the role of herbivorous fish in regulating algae dynamics through a mathematical modeling approach, considering three distinct scenarios. In the first scenario, they developed a mathematical model for a water ecosystem where oxygen levels are maintained by external sources. Expanding upon this framework, they introduced a second model in which oxygen levels are not externally regulated, allowing for a more dynamic interaction between oxygen availability and algae growth. Additionally, they extended their study to examine the effects of intensified algae mitigation efforts, assessing how these interventions influence ecosystem's stability and algae biomass dynamics. The proposed models illustrate

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significant shifts in the interactions between herbivorous fish and algae density under varying environmental conditions, offering valuable insights into the potential role of biological control strategies in managing HABs. Moreover, Sajid et al. [24] developed a mathematical model to examine the role of aquatic species, particularly fish population, in mitigating algal blooms. Their model analysis revealed significant shifts in ecosystem's dynamics, highlighting the potential effectiveness of fish-mediated strategies for controlling algae proliferation.

Although various types of algal blooms influence ecosystems and human life in multiple ways, HABs represent the most severe threat due to their detrimental ecological, economic, and public health consequences. Unlike non-toxic algal blooms, which may contribute to primary production and nutrient cycling, harmful algae are characterized by the excessive proliferation of toxin-producing algae species, leading to oxygen depletion, large-scale fish mortality, and biodiversity loss. To better understand and predict the dynamics of HABs under varying environmental and ecological conditions, some researchers have developed and analyzed mathematical models to investigate the factors influencing HAB formation, persistence, and decline. In this regard, the study of Yussuf et al. [25], a mathematical model with discrete delay was developed to describe HABs caused by *Pyrodinium bahamense*. The delay represents the time for toxin-producing phytoplankton (TPP) to mature and produce toxins, which accumulate in shellfish, posing human health risks. HABs reduce dissolved oxygen, causing fish mortality, while dinoflagellate cell walls can obstruct fish gills. The model shows that delay stabilizes the system, contrasting prior findings, and highlights the roles of nutrient concentration, dilution rate, and interspecies competition. A second model incorporating delay describes delayed fish mortality from toxin consumption. Nguyen-Quang et al. [26] developed a Turing instability-based model to analyze HAB patterns in lentic waters, demonstrating strong alignment with real-world observations. The model enhances understanding of bloom dynamics, focusing on biophysical thresholds and toxin quantification. Chakraborty et al. [27] utilized a mathematical modeling approach to compare the effects of three major anthropogenic factors—fertilizer application in agricultural fields, eutrophication from non-fertilizer sources, and overfishing on algal bloom dynamics and dissolved oxygen (DO) levels. Through a comprehensive sensitivity analysis, their study revealed that increased fertilizer usage in agriculture significantly accelerates algae growth and leads to a more rapid depletion of DO levels compared to eutrophication from other sources and overfishing. These findings highlight the critical role of agricultural run-off in driving HABs and emphasize the need for targeted nutrient management strategies to mitigate their ecological and environmental impacts. Thakur et al. [28] proposed a phytoplankton–zooplankton interaction model, incorporating non-toxic phytoplankton (NTP), TPP, and zooplankton dynamics using Holling type II and Monod–Haldane functional responses. The investigation highlights the role of diffusion in shaping algal bloom dynamics, as evidenced by the emergence of various Turing patterns. Furthermore, the study explores the destabilizing effect of time delay on stationary points, leading to the formation of limit cycles. It is observed that increasing the diffusion coefficient of zooplankton and time facilitates the transition of algal bloom patterns from a spot-strip mixture to isolated spot patterns, suggesting a potential mechanism for bloom stabilization. Belshiasheela et al. [29] investigated the dynamics of aquatic ecosystems involving non-toxic algae (a food source for zooplankton) and toxin-producing algae (which causes zooplankton mortality). The study examines how zooplankton predation influences algae density. A deterministic model is first developed, followed by a stochastic extension to assess its impact on population dynamics. The authors derived various equilibria, analyzed their local stability, and demonstrated the existence of Hopf bifurcation under specific parameter conditions. Numerical simulations compare deterministic and stochastic results, revealing that key parameters such as the growth rate of non-toxic algae and zooplankton predation rates significantly influence algal bloom dynamics. Minor adjustments to these parameters can increase non-toxic algae while reducing toxin-producing algae, aligning with policymakers' objectives for ecosystem management. These investigations have revealed multifaceted interactions between HABs and various ecosystem components, including their impacts on water chemistry, trophic relationships, and biodiversity patterns.

In this research article, we develop and analyze a comprehensive mathematical model to investigate the complex dynamics of interactions between HABs, fish, and humans. The model integrates the ecological processes governing HAB proliferation, the impact of HABs on fish population, and the potential risks posed to human health through contaminated water and seafood. By considering various biological and environmental factors, including nutrient dynamics, the growth of algae species, the response of fish, and human exposure pathways, we aim to better understand how these interconnected systems influence each other.

2 The mathematical model

In this section, we formulate a mathematical model to describe the interrelationships between HABs, dissolved nutrients, aquatic fauna (specifically fish population) within the water column and its impact on human population. The formulated model aims to characterize the complex interactions among these components, providing insights into the dynamics of aquatic ecosystems. To facilitate understanding and ensure clarity, we consider four dynamical variables, i.e., N (signifies nutrients), A (represents HABs), F (represents fish population), and H (signifies human population). For biological significance, we consider that the initial values of N , A , F , and H are non-negative, i.e., $N > 0$, $A \geq 0$, $F \geq 0$, $H \geq 0$. Thus, the state of space of proposed model system lies in $\mathbb{R}_+^4$.

In aquatic ecosystems, the growth of HABs is fundamentally regulated by the availability of dissolved nutrients, particularly phosphorus and nitrogen, which are essential for their proliferation. We assume that nutrients enter into the water body at a constant rate q , while natural depletion occurs proportional to nutrients present in the water body at a proportionality rate constant α_0 . Moreover, the consumption of nutrients by harmful algae is modeled by using a Michaelis–Menten type interaction, i.e., $\frac{\beta_1 N A}{\beta_{12} + \beta_{11} N}$,

where β_1 , β_{11} , and β_{12} describe the Michaelis–Menten kinetics. As harmful algae growth is directly linked to nutrient consumption; thus, we consider that HABs growth is proportional to this consumption rate, which is expressed as $\frac{\theta_1 \beta_1 N A}{\beta_{12} + \beta_{11} N}$, where θ_1 is a proportionality constant. Simultaneously, harmful algae undergoes natural depletion, which is represented by $\alpha_1 A$, where α_1 is the natural depletion rate. This process contributes to the nutrient cycle, as depleted algae are converted back into nutrients at a rate of $\pi_1 \alpha_1 A$, with π_1 being another proportionality constant. Combining the above-mentioned facts, the nutrients and HABs growth dynamics are represented by the following two differential equations:

$$\frac{dN}{dt} = q - \alpha_0 N - \beta_1 \frac{N A}{\beta_{12} + \beta_{11} N} + \pi_1 \alpha_1 A, \quad (1)$$

$$\frac{dA}{dt} = \theta_1 \beta_1 \frac{N A}{\beta_{12} + \beta_{11} N} - \alpha_1 A. \quad (2)$$

From Eq. (2), it is important to note that the condition $\theta_1 \beta_1 - \beta_{11} \alpha_1 > 0$ must hold to ensure the feasibility of Eq. (2).

Harmful algae play a complex role in aquatic ecosystems, with significant impact on fish population. Many fish species depend on algae as a primary food source, either directly by consuming algae or indirectly by feeding on organisms that rely on algae. This relationship is crucial for the growth, reproduction, and overall sustenance of fish population. However, the dynamics of this relationship can be drastically alter when HABs occur. The toxins released by harmful algae can directly poison fish, leading to mass mortality events. To model this fish-HABs interaction, we employ the Holling type II functional response, i.e., $\frac{k_1 A F}{m + A}$, k_1 represents the consumption rate of algae by fish population, and m is the half-saturation constant. This formulation captures the nonlinear nature of the consumption rate, which increases with algae density but asymptotically approaches a plateau. Moreover, the constant δ_0 represents the growth rate of the fish population under ideal conditions, where the absence of HABs allows the fish population to grow unhindered. However, when HABs are present, the decrease in the fish population due to the consumption of these toxic algae is assumed to be directly proportional to the amount of harmful algae consumed by the fish population. Thus, the term, $\left(\delta_0 - \frac{\lambda_1 k_1 A}{m + A}\right)$, represents the deformity in the fish population, where λ_1 is a proportionality constant that links the harmful effects of algae consumption to fish population growth. Consequently, the dynamics of fish population in the considered aquatic ecosystem can be represented by the following equation:

$$\frac{dF}{dt} = \left(\delta_0 - \frac{\lambda_1 k_1 A F}{m + A}\right) F - \delta_1 F^2. \quad (3)$$

Here, we consider δ_1 as the rate of decrease in fish population due to intra-specific competition. Also, the dynamics of harmful algae density in the presence of fish population can be written as

$$\frac{dA}{dt} = \theta_1 \beta_1 \frac{N A}{\beta_{12} + \beta_{11} N} - \alpha_1 A - \frac{k_1 A F}{m + A}. \quad (4)$$

The dynamics of the aquatic ecosystem is further complicated when human population reside near the water body. In our formulation, we represent human population growth using a logistic model, i.e., expressed as $r H \left(1 - \frac{H}{L}\right)$, where r is the intrinsic growth rate, and L is the carrying capacity of the environment. Consequently, human presence invariably leads to increased nutrient influx into the aquatic ecosystem. Thus, we model this anthropogenic nutrient input rate as pH , where p represents the anthropogenic nutrient input rate coefficient. Simultaneously, humans consume fish as their food source; thus, we employ a bilinear formulation $k_2 F H$, which represents the uptake rate of fish by humans, where k_2 is the fish uptake rate coefficient. Furthermore, the impact of fish consumption on human population growth is quantified by $\lambda_2 k_2 F H$, where λ_2 serves as a proportionality constant that converts consumed fish biomass into human population growth. This term encapsulates the nutritional benefit humans derive from fish consumption and its subsequent effect on population dynamics. Thus, the dynamics of human population near aquatic ecosystem can be written as

$$\frac{dH}{dt} = r H \left(1 - \frac{H}{L}\right) + \lambda_2 k_2 F H. \quad (5)$$

Moreover, in the presence of human population, the dynamics of nutrient input and fish population that are represented by Eqs. (1) and (3), respectively, are re-written as follows:

$$\frac{dN}{dt} = q + pH - \alpha_0 N - \frac{\beta_1 N A}{\beta_{12} + \beta_{11} N} + \pi_1 \alpha_1 A, \quad (6)$$

$$\frac{dF}{dt} = \left(\delta_0 - \frac{\lambda_1 k_1 A}{m + A}\right) F - \delta_1 F^2 - k_2 F H. \quad (7)$$

Thus, considering all the assumptions, we get the mathematical model that describes the nutrient–algae–fish interaction, when human population reside near the water body, i.e.,

$$\frac{dN}{dt} = \underbrace{q}_{\text{Nutrients influx}} + \underbrace{pH}_{\text{Anthropogenic nutrients influx}} - \underbrace{\alpha_0 N}_{\text{Natural depletion}} - \underbrace{\frac{\beta_1 N A}{\beta_{12} + \beta_{11} N}}_{\text{Nutrients uptake by algae}} + \underbrace{\pi_1 \alpha_1 A}_{\text{Growth due to algae depletion}},$$

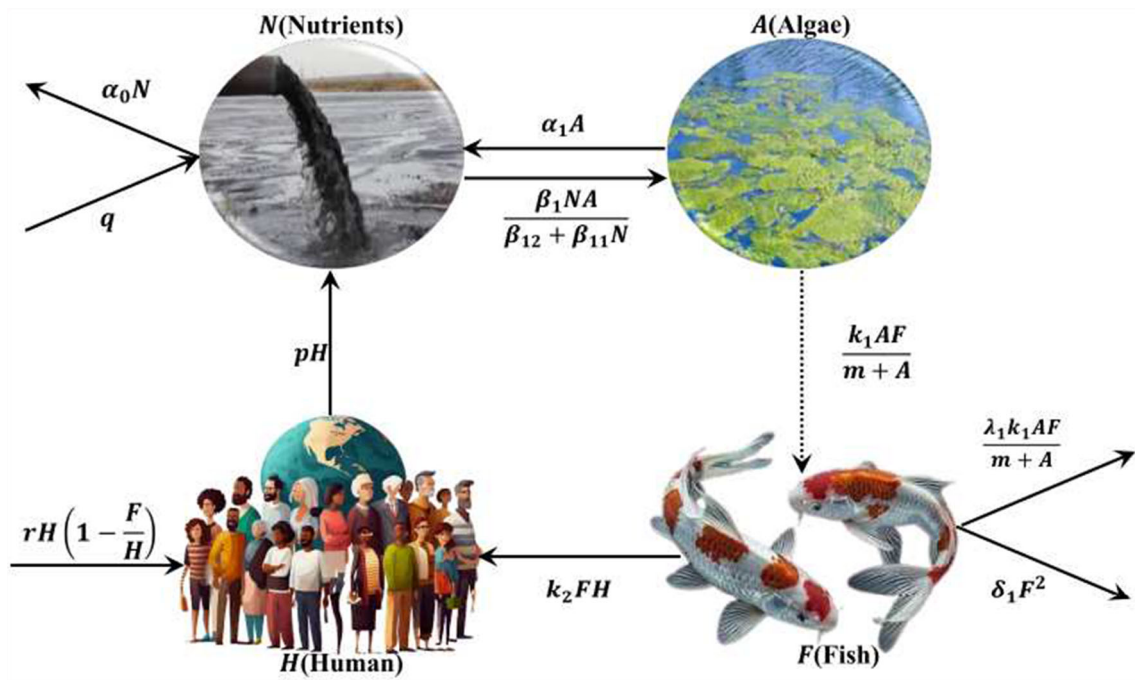


Fig. 1 Schematic diagram for model system (8)

$$\begin{aligned}
 \frac{dA}{dt} &= \underbrace{\theta_1 \beta_1 \frac{NA}{\beta_{12} + \beta_{11} N}}_{\text{Growth due to nutrients uptake}} - \underbrace{\alpha_1 A}_{\text{Natural depletion}} - \underbrace{\frac{k_1 AF}{m + A}}_{\text{Algae uptake by fish}}, \\
 \frac{dF}{dt} &= \underbrace{\left(\delta_0 - \frac{\lambda_1 k_1 A}{m + A} \right) F}_{\text{Deformity due to uptake of HABs}} - \underbrace{\delta_1 F^2}_{\text{Intra-sepecific competition}} - \underbrace{k_2 FH}_{\text{Fish uptake by humans}}, \\
 \frac{dH}{dt} &= \underbrace{rH \left(1 - \frac{H}{L} \right)}_{\text{Growth of human population}} + \underbrace{\frac{\lambda_2 k_2 FH}{m + A}}_{\text{Growth due to fish uptake}}.
 \end{aligned} \tag{8}$$

The region of attraction, containing all solutions that start within the positive orthant, is enclosed in

$$\Omega = \left\{ (N, A, F, H) : 0 < N + A \leq \frac{q + \frac{pL}{r} \left(r + \frac{\lambda_2 k_2 \delta_0}{\delta_1} \right)}{Q_m}, F(0) \leq F \leq \frac{\delta_0}{\delta_1}, 0 < H \leq \frac{L}{r} \left(r + \frac{\lambda_2 k_2 \delta_0}{\delta_1} \right) \right\},$$

where $Q_m = \min\{\alpha_0, (1 - \pi_1)\alpha_1\}$.

The schematic diagram of the model system (8) is shown in Fig. 1, and the descriptions of all parameters are provided in Table 1.

The model system (8) illustrates the interconnected dynamics of an aquatic ecosystem, where nutrients fuel HABs, which sustain and harm fish populations through consumption and toxicity. Fish populations, in turn, support human populations through consumption, while humans contribute to nutrient influx, creating a feedback loop. The system captures the balance between natural processes and anthropogenic influences, highlighting how human activities can disrupt ecological stability, leading to HAB proliferation, fish population decline, and alter nutrient cycles. It serves as a tool to analyze and mitigate the impacts of human intervention on aquatic ecosystems.

3 Existence and local stability of equilibrium

This section is dedicated to a comprehensive analysis of the model system (8). Our investigation begins with a thorough examination of both the boundary and interior equilibria of the system. These equilibria represent steady states, where the system's variables remain constant over time, providing crucial insights into the long-term behavior of the model. Following this, we delve into a local stability analysis of all the equilibria we have identified. This two-step approach—first identifying the equilibria and then analyzing

Table 1 Description of considered parameters and their units in model system (8)

Parameter	Description	Unit
q	Influx rate of nutrients into the water body	$\mu\text{g liter}^{-1}\text{day}^{-1}$
p	Influx of nutrients into the water body due to anthropogenic activity	$\mu\text{g liter}^{-1}\text{person}^{-1}\text{day}^{-1}$
α_0	Natural depletion rate of nutrients	day^{-1}
β_1	Maximum uptake rate of nutrients by HABs	day^{-1}
β_{12}	Half-saturation constant of nutrients	$\mu\text{g liter}^{-1}$
β_{11}	Proportionality constant describing Michaelis–Menten interaction	No unit
α_1	Natural depletion of HABs	day^{-1}
π_1	Proportionality constant that represent HAB biomass conversion into nutrient	No unit
θ_1	Proportionality constant that represents growth of HABs by absorbing nutrients	No unit
k_1	Uptake rate of HABs by fish population	$\mu\text{g liter}^{-1}\text{fish}^{-1}\text{day}^{-1}$
δ_0	Growth rate of fish population	day^{-1}
m	Half-saturation constant of HABs	$\mu\text{g liter}^{-1}$
λ_1	Proportionality constant that quantifies the reduction in fish population caused by the consumption of HABs	No unit
δ_1	Death rate of fish population due to intra-specific competition	$\text{fish}^{-1}\text{day}^{-1}$
k_2	Consumption rate of fish by human population	$\text{person}^{-1}\text{day}^{-1}$
λ_2	Proportionality constant that represent growth of human population due to consumption of fish	No unit
r	Intrinsic growth rate of human population	day^{-1}
L	Carrying capacity of human population	person

their stability, allows us to build a complete picture of the system's dynamics, revealing potential steady states and their robustness to small disturbances.

3.1 Boundary equilibrium

Boundary equilibria represent states, where one or more variables of the system are absent. The model system (8) exhibits seven feasible boundary equilibrium. The existence of boundary equilibria are discussed below:

- (i) The equilibrium $E_1\left(\frac{q}{\alpha_0}, 0, 0, 0\right)$ always exists.
- (ii) If $F = H = 0$, then from second equation model system (8), we have $N = \frac{\beta_{12}\alpha_1}{\theta_1\beta_1 - \beta_{11}\alpha_1}$ (say N_2). Thus, from first equation of model system (8), we have $A = \frac{\alpha_0\left(\frac{q}{\alpha_0} - N_2\right)}{\left(\frac{\beta_1 N_2}{\beta_{12} + \beta_{11}N_2} - \pi_1\alpha_1\right)}$ (say A_2). Therefore, we can say that equilibrium $E_2(N_2, A_2, 0, 0)$ is feasible provided $N_2 < \frac{q}{\alpha_0}$ and $\beta_1 N_2 > \pi_1\alpha_1(\beta_{12} + \beta_{11}N_2)$.
- (iii) If $A = H = 0$, then from first and third equation of model system (8), we have $N = \frac{q}{\alpha_0}$ and $F = \frac{\delta_0}{\delta_1}$. Thus, the equilibrium $E_3\left(\frac{q}{\alpha_0}, 0, \frac{\delta_0}{\delta_1}, 0\right)$ always exists.
- (iv) If only $H = 0$, then from first three equations of model system (8), we have

$$q - \alpha_0 N - \frac{\beta_1 N A}{\beta_{12} + \beta_{11} N} + \pi_1 \alpha_1 A = 0, \quad (9)$$

$$\frac{\theta_1 \beta_1 N}{\beta_{12} + \beta_{11} N} - \alpha_1 - \frac{k_1 F}{m + A} = 0, \quad (10)$$

$$\delta_0 - \frac{\lambda_1 k_1 A}{m + A} - \delta_1 F = 0. \quad (11)$$

From Eq. (11), we have

$$F = \frac{1}{\delta_1(m + A)}(\delta_0 m + (\delta_0 - \lambda_1 k_1)A), \text{ here } F > 0 \text{ if } \delta_0 > \lambda_1 k_1. \quad (12)$$

Substituting the value of F from Eq. (12) in Eq. (10), we have

$$\frac{\theta_1 \beta_1 N}{\beta_{12} + \beta_{11} N} - \alpha_1 - \frac{k_1}{\delta_1(m + A)^2}(\delta_0 m + (\delta_0 - \lambda_1 k_1)A) = 0. \quad (13)$$

Table 2 Obtained boundary equilibria and their existence condition(s)

Obtained boundary equilibrium	Existence condition
HAB-fish-human free equilibrium $E_1\left(\frac{q}{\alpha_0}, 0, 0, 0\right)$	Always exists
Fish-human free equilibrium $E_2(N_2, A_2, 0, 0)$, here $N_2 = \frac{\beta_{12}\alpha_1}{\theta_1\beta_1 - \beta_{11}\alpha_1} \quad A_2 = \frac{\alpha_0\left(\frac{q}{\alpha_0} - N_2\right)}{\left(\frac{\beta_1 N_2}{\beta_{12} + \beta_{11}N_2} - \pi_1\alpha_1\right)}$	Feasible if $N_2 < \frac{q}{\alpha_0}$ and $\beta_1 N_2 > \pi_1\alpha_1$ $(\beta_{12} + \beta_{11}N_2)$
HAB-human free equilibrium $E_3\left(\frac{q}{\alpha_0}, 0, \frac{\delta_0}{\delta_1}, 0\right)$	Always exists
Human free equilibrium $E_4(N_4, A_4, F_4, 0)$	Exists if $\delta_0 > \lambda_1 k_1$
HAB-fish free equilibrium $E_5\left(\frac{q+pL}{\alpha_0}, 0, 0, L\right)$	Always exists
HAB free equilibrium $E_6\left(\frac{q+pH_6}{\alpha_0}, 0, F_6, H_6\right)$	Feasible if $\delta_0 > k_2 L$
Fish free equilibrium $E_7(N_7, A_7, 0, L)$, where $N_7 = \frac{\beta_{12}\alpha_1}{\theta_1\beta_1 - \beta_{11}\alpha_1} \quad \text{and} \quad A_7 = \frac{\alpha_0\left(\frac{q+pL}{\alpha_0} - N_7\right)}{\left(\frac{\beta_1 N_7}{\beta_{12} + \beta_{11}N_7} - \pi_1\alpha_1\right)}$	Feasible if $N_7 < \frac{q+pL}{\alpha_0}$ and $\beta_1 N_7 > \pi_1\alpha_1$ $(\beta_{12} + \beta_{11}N_7)$

From Eq. (13), it is apparent that (a) if $A = 0$, we have $N = \frac{\beta_{12}(\alpha_1 + \frac{\delta_0 k_1}{\delta_1 m})}{\theta_1 \beta_1 - \beta_{11}(\alpha_1 + \frac{\delta_0 k_1}{\delta_1 m})}$, (b) if $N = 0$, we have a quadratic equation in A , i.e.,

$$\alpha_1 A^2 + \left(2\alpha_1 m + \frac{k_1}{\delta_0}(\delta_0 - \lambda_1 k_1)\right)A + \alpha_1 m^2 + k_1 m \frac{\delta_0}{\delta_1} = 0. \quad (14)$$

Thus, Eq. (14) has no positive root if $\delta_0 > \lambda_1 k_1$. (c) $\frac{\theta_1 \beta_1 \beta_{12}}{\beta_{12} + \beta_{11}N} \frac{dN}{dA} = -\frac{k_1}{\delta_1(m+A)^3}(m(\delta_0 + \lambda_1 k_1) + (\delta_0 - \lambda_1 k_1)A)$. Thus, the isocline curve in $A - N$ plane is decreasing if $\delta_0 > \lambda_1 k_1$. (d) $N = \frac{\alpha_1 \beta_{12}}{\theta_1 \beta_1 - \alpha_1 \beta_{11}}$ is an asymptote. Further, from Eq. (9), we can note that (a) if $A = 0$, we have $N = \frac{q}{\alpha_0}$, (b) if $N = 0$, we have $A = -\frac{q}{\pi_1 \alpha_1}$, (c) $\frac{dN}{dA} = (-q + \alpha_0 N) / \left[A\left(\alpha_0 + \frac{\beta_1 \beta_{12} A}{(\beta_{12} + \beta_{11}N)^2}\right)\right] < 0$ as $N < \frac{q}{\alpha_0}$. and (a) $N = \frac{\pi_1 \beta_{12} \alpha_1}{\beta_1 - \pi_1 \beta_{11} \alpha_1}$ is an asymptote. Thus, we can say that isoclines (9) and (13) can intersect at no or one point in the positive quadrant of $A - N$ plane. Thus, we can say that if $\delta_0 > \lambda_1 k_1$, then equilibrium $E_4(N_4, A_4, F_4, 0)$ of model system (8) either exists uniquely or in some cases may not exist.

- (v) If $F = A = 0$, then from third equation of model system (8), we have $H = L$ and thus from first equation of model system (8), we have $N = \frac{q+pL}{\alpha_0}$. Thus, equilibrium $E_5\left(\frac{q+pL}{\alpha_0}, 0, 0, L\right)$ always exists.
- (vi) If only $A = 0$, from fourth equation of model system (8), we have $H = \frac{L}{r}(\lambda_2 k_1 F + r)$. Thus, using this value of H in third equation of model system (8), we have $F = \frac{r(\delta_0 - k_2 L)}{r\delta_1 + k_2^2 L \lambda_2}$ (say F_6). Here F_6 is positive only if $\delta_0 > k_2 L$. Now, using $F = F_6$ in third equation of model system (8), we have $H = \lambda_2 k_2 L \left(\frac{(\delta_0 - k_2 L)}{r\delta_1 + k_2^2 L \lambda_2}\right) + L$ (say H_6). Again using $H = H_6$, we have $N = \frac{1}{\alpha_0} \left[q + p \left(\lambda_2 k_2 L \left(\frac{(\delta_0 - k_2 L)}{r\delta_1 + k_2^2 L \lambda_2}\right) + L\right)\right]$ (say N_6). Thus, we can say that equilibrium $E_6\left(\frac{q+pH_6}{\alpha_0}, 0, F_6, H_6\right)$ exists only if $\delta_0 > k_2 L$.
- (vii) If only $F = 0$, then from second and fourth equations of model system (8), we have $N = \frac{\beta_{12}\alpha_1}{\theta_1\beta_1 - \beta_{11}\alpha_1}$ (say N_7) and $H = L$, respectively. Thus, using $N = N_7$ in first equation of model system (8), we have $A = \frac{\alpha_0\left(\frac{q+pL}{\alpha_0} - N_7\right)}{\left(\frac{\beta_1 N_7}{\beta_{12} + \beta_{11}N_7} - \pi_1\alpha_1\right)}$ (say A_7). Thus, we can say that equilibrium $E_7(N_7, A_7, 0, L)$ exists, provided $N_7 < \frac{q+pL}{\alpha_0}$ and $\beta_1 N_7 - \pi_1\alpha_1(\beta_{12} + \beta_{11}N_7) > 0$.

All the seven boundary equilibria and their existence condition(s) are summarized in Table 2.

Theorem 1 For the model system (8), the boundary equilibrium

- (i) E_i ($i = 1, 2, \dots, 4$) are unstable, whenever exists.
- (ii) $E_5\left(\frac{q+pL}{\alpha_0}, 0, 0, L\right)$ is stable if $\delta_0 < k_2 L$ and $\theta_1 \beta_1 (q + pL) - \alpha_1 (\beta_{12} \alpha_0 + \beta_{11} (q + pL)) < 0$.
- (iii) $E_6\left(\frac{q+pH_6}{\alpha_0}, 0, F_6, H_6\right)$ is stable if $\theta_1 \beta_1 (q + pH_6) - \left(\alpha_1 + \frac{k_1 F_6}{m}\right) (\beta_{12} \alpha_0 + \beta_{11} (q + pH_6)) < 0$.
- (iv) The boundary $E_7(N_7, A_7, 0, L)$ is stable if $\mathcal{B}_j > 0$ for ($j = 1, 3, 4$) and $\mathcal{B}_3(\mathcal{B}_1 \mathcal{B}_2 - \mathcal{B}_3) - \mathcal{B}_1^2 \mathcal{B}_4 > 0$.

Proof To discuss the stability of obtained boundary equilibria, we first evaluate the Jacobian matrix of model system (8), i.e.,

$$J = \begin{bmatrix} -a_{11} & -a_{22} & 0 & p \\ a_{21} & a_{22} & -a_{23} & 0 \\ 0 & -a_{32} & a_{33} & -k_2 F \\ 0 & 0 & \lambda_2 k_2 H & a_{44} \end{bmatrix},$$

where $a_{11} = \alpha_0 + \frac{\beta_1 \beta_{12} A}{(\beta_{12} + \beta_{11} N)^2}$, $a_{12} = \frac{\beta_1 N}{(\beta_{12} + \beta_{11} N)} + \pi_1 \alpha_1$, $a_{21} = \frac{\theta_1 \beta_1 \beta_{12} A}{(\beta_{12} + \beta_{11} N)^2}$, $a_{22} = \frac{\theta_1 \beta_1 N}{(\beta_{12} + \beta_{11} N)} - \alpha_1 - \frac{k_1 m F}{(m+A)^2}$, $a_{23} = \frac{k_1 A}{m+A}$, $a_{32} = \frac{\lambda_1 k_1 m F}{(m+A)^2}$, $a_{33} = \delta_0 - \frac{\lambda_1 k_1 A}{m+A} - 2\delta_1 F - k_2 H$, $a_{44} = r(1 - \frac{2H}{L}) + \lambda_2 k_2 F$ at each equilibria..

- (i) Now, the eigenvalues of Jacobian matrix J , evaluated at equilibrium E_1 , are $\Psi_{11} = -\alpha_0$, $\Psi_{12} = \left(\frac{\theta_1 \beta_1 q}{\alpha_0 \beta_{12} + q \beta_{11}} - \alpha_1\right)$, $\Psi_{13} = \delta_0$ and $\Psi_{14} = r$. The two eigenvalues Ψ_{13} and Ψ_{14} of Jacobian matrix evaluated at E_1 are always positive. Consequently, we can say that equilibrium $E_1\left(\frac{q}{\alpha_0}, 0, 0, 0\right)$ is unstable, whenever exists. Jacobian matrix J evaluated at equilibrium E_2 has two eigenvalues $\Psi_{21} = r$, $\Psi_{22} = \delta_0 - \frac{\lambda_1 k_1 A_2}{m+A_2}$ and other two eigenvalues Ψ_{23}, Ψ_{24} are solutions of following quadratic equation:

$$\Psi^2 + \left(\alpha_0 + \frac{\beta_1 \beta_{12} A_2}{(\beta_{12} + \beta_{11} N_2)^2}\right) \Psi + \frac{\theta_1 \beta_1 \beta_{12} A_2}{(\beta_{12} + \beta_{11} N_2)^2} \left(\frac{\beta_1 N_2}{(\beta_{12} + \beta_{11} N_2)} - \pi_1 \alpha_1\right) = 0.$$

As one eigenvalue, i.e., $\Psi_{21} = r$ is always positive. Thus, the equilibrium $E_2(N_2, A_2, 0, 0)$ is unstable, whenever exists. The Jacobian matrix J , when evaluated at equilibrium E_3 has eigenvalues $\Psi_{31} = -\alpha_0$, $\Psi_{32} = \frac{\theta_1 \beta_1 q}{\alpha_0 \beta_{12} + q \beta_{11}} - \alpha_1 - \frac{\delta_0 k_1}{\delta_1 m}$, $\Psi_{33} = -\delta_0$, and $\Psi_{34} = r + \lambda_2 k_2 \frac{\delta_0}{\delta_1}$. As the fourth eigenvalue Ψ_{34} is always positive; thus, we can say that equilibrium $E_3\left(\frac{q}{\alpha_0}, 0, \frac{\delta_0}{\delta_1}, 0\right)$ is unstable, whenever exists. Now, the matrix J , when evaluated at E_4 , gives two eigenvalues as $\Psi_{41} = -\delta_1 F_4$, $\Psi_{42} = r + \lambda_2 k_2 F_4$ and other two eigenvalues Ψ_{43}, Ψ_{44} are the solution of following quadratic equation:

$$\Psi^2 + \left(\alpha_0 + \frac{\beta_1 \beta_{12} A_4}{(\beta_{12} + \beta_{11} N_4)^2} - \frac{k_1 A_4 F_4}{(m + A_4)^2}\right) \Psi + \frac{\beta_1 \beta_{12} A_4}{(\theta_1 \beta_{12} + \beta_{11} N_4)^2} \left(\frac{\beta_1 N_4}{\beta_{12} + \beta_{11} N_4} - \pi_1 \alpha_1\right) = 0.$$

Here, one eigenvalue Ψ_{42} is always positive; thus, we can say that $E_4(N_4, A_4, F_4, 0)$ is unstable, whenever exists.

- (ii) At equilibrium E_5 , Jacobian matrix J has two eigenvalues $\Psi_{51} = -\alpha_0$, $\Psi_{52} = \frac{\theta_1 \beta_1 (q+pL)}{(\beta_{12} \alpha_0 + \beta_{11} (q+pL))} - \alpha_1$, and other two eigenvalues Ψ_{53}, Ψ_{54} are the solution of following quadratic equation:

$$\Psi^2 - (\delta_0 - k_2 L - r) \Psi - r(\delta_0 - k_2 L) + \lambda_2 k_2^2 L = 0.$$

Therefore, if $[\theta_1 \beta_1 (q + pL) - \alpha_1 (\beta_{12} \alpha_0 + \beta_{11} (q + pL))] < 0$, then, applying the Routh–Hurwitz criterion, we can conclude that the equilibrium $E_5\left(\frac{q+pL}{\alpha_0}, 0, 0, L\right)$ is locally asymptotically stable, provided the condition $(\delta_0 - k_2 L) < 0$ is satisfied.

- (iii) Jacobian matrix J (evaluated at equilibrium E_6) has two eigenvalues $\Psi_{61} = -\alpha_0$, $\Psi_{62} = \frac{\theta_1 \beta_1 (q+pH_6)}{\beta_{12} \alpha_0 + \beta_{11} (q+pH_6)} - \alpha_1 - \frac{k_1 F_6}{m}$ and other two eigenvalues Ψ_{63}, Ψ_{64} are the solution of following quadratic equation:

$$\Psi^2 + \left(\delta_1 F_6 + \frac{r H_6}{L}\right) \Psi + \frac{r \delta_1 F_6 H_6}{L} + \lambda_2 k_2^2 H_6 = 0. \quad (15)$$

Since $\left(\delta_1 F_6 + \frac{r H_6}{L}\right) > 0$ and $\left(\frac{r \delta_1 F_6 H_6}{L} + \lambda_2 k_2^2 H_6\right) > 0$; thus, using the Routh–Hurwitz criterion, we can say that both roots of Eq. (15) are either negative or contain the negative real parts. Therefore, equilibrium $E_6\left(\frac{q+pH_6}{\alpha_0}, 0, F_6, H_6\right)$ is locally asymptotically stable if

$$\left[\theta_1 \beta_1 (q + pH_6) - \left(\alpha_1 + \frac{k_1 F_6}{m}\right) (\beta_{12} \alpha_0 + \beta_{11} (q + pH_6))\right] < 0,$$

otherwise unstable.

- (iv) Eigenvalues of matrix J (evaluated at E_7) are the solutions of following algebraic equation:

$$\Psi^4 + \mathcal{B}_1 \Psi^3 + \mathcal{B}_2 \Psi^2 + \mathcal{B}_3 \Psi + \mathcal{B}_4 = 0,$$

where

$$\begin{aligned} \mathcal{B}_1 &= r + \alpha_0 + \frac{\beta_1 \beta_{12} A_7}{(\beta_{12} + \beta_{11} N_7)^2} - (\delta_0 - k_2 L) + \frac{\lambda_1 k_1 A_7}{m + A_7}, \\ \mathcal{B}_2 &= -r(\delta_0 - k_2 L) + \frac{\lambda_1 r k_1 A_7}{m + A_7} + \lambda_2 k_2^2 L^2 + \left(\alpha_0 + \frac{\beta_1 \beta_{12} A_7}{(\beta_{12} + \beta_{11} N_7)^2}\right) \left(r - (\delta_0 - k_2 L) + \frac{\lambda_1 k_1 A_7}{m + A_7}\right) \\ &\quad + \frac{\theta_1 \beta_1 \beta_{12} A_7}{(\beta_{12} + \beta_{11} N_7)^2} \left(\frac{\beta_1 N_7}{\beta_{12} + \beta_{11} N_7} - \pi_1 \alpha_1\right), \end{aligned}$$

$$\begin{aligned} \mathcal{B}_3 &= r \left(\alpha_0 + \frac{\beta_1 \beta_{12} A_7}{(\beta_{12} + \beta_{11} N_7)^2} \right) \left(-(\delta_0 - k_2 L) + \frac{\lambda_1 k_1 A_7}{m + A_7} \right) + \frac{\theta_1 \beta_1 \beta_{12} A_7}{(\beta_{12} + \beta_{11} N_7)^2} \left(\frac{\beta_1 N_7}{\beta_{12} + \beta_{11} N_7} - \pi_1 \alpha_1 \right) \\ &\quad \times \left(r - (\delta_0 - k_2 L) + \frac{\lambda_1 k_1 A_7}{m + A_7} \right), \\ \mathcal{B}_4 &= \frac{\theta_1 \beta_1 \beta_{12} A_7}{(\beta_{12} + \beta_{11} N_7)^2} \left(\frac{\beta_1 N_7}{\beta_{12} + \beta_{11} N_7} - \pi_1 \alpha_1 \right) \left(-r(\delta_0 - k_2 L) - \frac{\lambda_1 r k_1 A_7}{m + A_7} + \lambda_2 k_2^2 L^2 \right). \end{aligned}$$

Thus, using Routh–Hurwitz criterion [30], we can say that equilibrium $E_7(N_7, A_7, 0, L)$ is locally asymptotically stable if $\mathcal{B}_j > 0$ ($j = 1, 3, 4$) and $\mathcal{B}_3(\mathcal{B}_1 \mathcal{B}_2 - \mathcal{B}_3) - \mathcal{B}_1^2 \mathcal{B}_4 > 0$, otherwise unstable. $\square$

3.2 Interior equilibrium

An interior equilibrium of the model system (8) represents a steady-state solution where all dynamical variables remain strictly positive over time. Unlike other equilibria, where one or more dynamic variables may vanish, the interior equilibrium reflects a state of coexistence among all components of the system. The existence of such an equilibrium is crucial for understanding the long-term dynamics and stability of the system (8).

Theorem 2 *The model system (8) may exhibit no, one or two interior equilibrium $E^*(N^*, A^*, F^*, H^*)$ if $\mathfrak{B} > \lambda_1 k_1$, where $\mathfrak{B} = \delta_0 - k_2 L$.*

Proof The equilibrium E^* can be obtained by analyzing following algebraic equations:

$$q + pH - \alpha_0 N - \frac{\beta_1 N A}{\beta_{12} + \beta_{11} N} + \pi_1 \alpha_1 A = 0, \quad (16)$$

$$\frac{\theta_1 \beta_1 N}{\beta_{12} + \beta_{11} N} - \alpha_1 - \frac{k_1 F}{m + A} = 0, \quad (17)$$

$$\delta_0 - \frac{\lambda_1 k_1 A}{m + A} - \delta_1 F - k_2 H = 0, \quad (18)$$

$$r \left(1 - \frac{H}{L} \right) + \lambda_2 k_2 F = 0. \quad (19)$$

Now, from Eq. (19), we have

$$H = \frac{L}{r} (r + \lambda_2 k_2 F). \quad (20)$$

Substituting the value of H from Eq. (20) in Eq. (18) and consider $\mathfrak{B} = \delta_0 - k_2 L$, we have

$$F = \frac{r}{(\delta_1 r + \lambda_2 k_2^2 L)(m + A)} (\mathfrak{B} m + (\mathfrak{B} - \lambda_1 k_1) A). \quad (21)$$

Also, substituting the value of F from Eq. (21) in Eq. (20), we have

$$H = L \left[\frac{\lambda_2 k_2 (\mathfrak{B} m + (\mathfrak{B} - \lambda_1 k_1) A)}{(\delta_1 r + \lambda_2 k_2^2 L)(m + A)} + 1 \right]. \quad (22)$$

From Eqs. (21) and (22), we can say that $F > 0$ and $H > 0$, if $\mathfrak{B} > \lambda_1 k_1$. Now, using the value of F from Eq. (21) in Eq. (17), we have an equation in N and A , i.e.,

$$\frac{\theta_1 \beta_1 N}{\beta_{12} + \beta_{11} N} - \alpha_1 - \frac{r k_1}{(\delta_1 r + \lambda_2 k_2^2 L)(m + A)^2} (\mathfrak{B} m + (\mathfrak{B} - \lambda_1 k_1) A) = 0. \quad (23)$$

From Eq. (23), it is apparent that (a) if $N = 0$, we have a quadratic equation in A , i.e., $\alpha_1 (\delta_1 r + \lambda_2 k_2^2 L)(m + A)^2 + k_1 r (\mathfrak{B} - \lambda_1 k_1) A + \mathfrak{B} k_1 = 0$, which has no positive root if $\mathfrak{B} > \lambda_1 k_1$, (b) if $A = 0$, we have

$N = \frac{\beta_{12} \left(\alpha_1 + \frac{k_1 r \mathfrak{B}}{m(\delta_1 r + \lambda_2 k_2^2 L)} \right)}{\theta_1 \beta_1 - \beta_{11} \left(\alpha_1 + \frac{k_1 r \mathfrak{B}}{m(\delta_1 r + \lambda_2 k_2^2 L)} \right)}$, (c) $N = \frac{\alpha_1 \beta_{12}}{\theta_1 \beta_1 - \alpha_1 \beta_{11}}$ is an asymptote, and (d) $\frac{\theta_1 \beta_1 \beta_{12} N}{(\beta_{12} + \beta_{11})^2} \frac{dN}{dA} = -\frac{k_1 r}{(\delta_1 r + \lambda_2 k_2^2 L)(\mathfrak{B} - \lambda_1 k_1)(m + A)^3} \left[A + \frac{m(\mathfrak{B} + \lambda_1 k_1)}{\mathfrak{B} - \lambda_1 k_1} \right]$, which implies that $dN/dA < 0$ if $\mathfrak{B} > \lambda_1 k_1$. Now, if we substitute the value of H in Eq. (16), we have another equation in N and A , i.e.,

$$q + pL \left[\frac{\lambda_2 k_2 (\mathfrak{B} m + (\mathfrak{B} - \lambda_1 k_1) A)}{(\delta_1 r + \lambda_2 k_2^2 L)(m + A)} + 1 \right] - \alpha_0 N - \frac{\beta_1 N A}{\beta_{12} + \beta_{11} N} + \pi_1 \alpha_1 A = 0. \quad (24)$$

From Eq. (24), it is apparent that (a) if $N = 0$, we have quadratic equation in A , i.e.,

$$\pi_1 \alpha_1 A^2 + \left(q + pL + \pi_1 \alpha_1 m + \frac{\lambda_2 k_2 pL}{\delta_1 r + \lambda_2 k_2^2 L} (\mathfrak{B} - \lambda_1 k_1) \right) A + mq + mpL + \frac{\lambda_2 k_2 \mathfrak{B} mpL}{\delta_1 r + \lambda_2 k_2^2 L} = 0,$$

which has no positive root if $\mathfrak{B} > \lambda_1 k_1$. (b) If $A = 0$, we have $N = \frac{1}{\alpha_0} \left(q + pL + \frac{\lambda_2 k_2 \mathfrak{B} pL}{\delta_1 r + \lambda_2 k_2^2 L} \right)$, (c) $N = \frac{\pi_1 \alpha_1 \beta_{12}}{\beta_{11} - \pi_1 \alpha_1 \beta_{11}}$ is an asymptote of (24), (d) $\left(\alpha_0 + \frac{\beta_1 \beta_{12} A}{(\beta_{12} + \beta_{11} N)^2} \right) \frac{dN}{dA} = \alpha_0 \left(N - \frac{q+pH}{\alpha_0} \right) - pL \cdot \frac{\lambda_2 k_2 \lambda_1 k_1 m}{(\delta_1 r + \lambda_2 k_2^2 L)(m+A)^2} < 0$, if $N < \frac{q+pH}{\alpha_0}$. Thus, we can say that isoclines (23) and (24) may intersect at no, one or two points in the positive quadrant of $A - N$ plane. Consequently, model system (8) may have no, one or two interior equilibrium $E^*(N^*, A^*, F^*, H^*)$ if $\mathfrak{B} > \lambda_1 k_1$. $\square$

Remark 1 For the existence of interior equilibrium, the condition $\delta_0 - k_2 L > \lambda_1 k_1$ must hold. This ensures that the fish population's intrinsic growth rate (δ_0) compensates for both the maximum harvesting pressure from humans ($k_2 L$) and the toxic effects of HABs ($\lambda_1 k_1$). Violation of this inequality would lead to the collapse of the fish population due to combined anthropogenic and ecological stressors. The condition thus serves as a critical threshold for sustainable ecosystem management, emphasizing the need to balance human activities with HAB mitigation to maintain viable fish stocks.

Theorem 3 The interior equilibrium $E^*(N^*, A^*, F^*, H^*)$ is locally asymptotically stable if $C_j > 0$ for $(j = 1, 3, 4)$ and $C_3(C_1 C_2 - C_3) - C_1^2 C_4 > 0$.

Proof The Jacobian matrix for model system (8) that is evaluated at equilibrium E^* is given as

$$J^* = \begin{bmatrix} -a_{11} & -a_{12} & 0 & p \\ a_{21} & a_{22} & -a_{23} & 0 \\ 0 & -a_{32} & -\delta_1 F^* & -k_2 F^* \\ 0 & 0 & \lambda_2 k_2 H^* & -\frac{rH^*}{L} \end{bmatrix},$$

where

$$\begin{aligned} a_{11} &= \alpha_0 + \frac{\beta_1 \beta_{12} A^*}{(\beta_{12} + \beta_{11} N^*)^2}, \quad a_{12} = \frac{\beta_1 N^*}{\beta_{12} + \beta_{11} N^*} - \pi_1 \alpha_1, \quad a_{21} = \frac{\theta_1 \beta_1 \beta_{12} A^*}{(\beta_{12} + \beta_{11} N^*)^2}, \\ a_{22} &= \frac{k_1 A^* F^*}{(m + A^*)^2}, \quad a_{23} = \frac{k_1 A^*}{m + A^*}, \quad a_{32} = \frac{\lambda_1 k_1 m F^*}{(m + A^*)^2}. \end{aligned}$$

The eigenvalues of matrix J^* are given by the following characteristic equation:

$$\Psi^4 + C_1 \Psi^3 + C_2 \Psi^2 + C_3 \Psi + C_4 = 0, \quad (25)$$

where

$$\begin{aligned} C_1 &= \delta_1 F^* + \frac{rH^*}{L} + \alpha_0 + \frac{\beta_1 \beta_{12} A^*}{(\beta_{12} + \beta_{11} N^*)^2} - \frac{k_1 A^* F^*}{(m + A^*)^2}, \\ C_2 &= \frac{r\delta_1 F^* H^*}{L} + \left(\delta_1 F^* + \frac{rH^*}{L} \right) \left(\alpha_0 + \frac{\beta_1 \beta_{12} A^*}{(\beta_{12} + \beta_{11} N^*)^2} - \frac{k_1 A^* F^*}{(m + A^*)^2} \right) - \frac{k_1 A^* F^*}{(m + A^*)^2} \left(\alpha_0 + \frac{\beta_1 \beta_{12} A^*}{(\beta_{12} + \beta_{11} N^*)^2} \right) \\ &\quad + \lambda_2 k_2^2 H^{*2} + \frac{\theta_1 \beta_1 \beta_{12} A^*}{(\beta_{12} + \beta_{11} N^*)^2} \left(\frac{\beta_1 N^*}{\beta_{11} + \beta_{12} N^*} - \pi_1 \alpha_1 \right), \\ C_3 &= \frac{r\delta_1 F^* H^*}{L} \left(\alpha_0 + \frac{\beta_1 \beta_{12} A^*}{(\beta_{12} + \beta_{11} N^*)^2} - \frac{k_1 A^* F^*}{(m + A^*)^2} \right) - \frac{k_1 A^* F^*}{(m + A^*)^2} \left(\alpha_0 + \frac{\beta_1 \beta_{12} A^*}{(\beta_{12} + \beta_{11} N^*)^2} \right) - \frac{\lambda_1 k_1^2 m A^* F^*}{(m + A^*)^3} \\ &\quad + \lambda_2 k_2^2 F^* H^* \left(\alpha_0 + \frac{\beta_1 \beta_{12} A^*}{(\beta_{12} + \beta_{11} N^*)^2} - \frac{k_1 A^* F^*}{(m + A^*)^2} \right) + \frac{\theta_1 \beta_1 \beta_{12} A^*}{(\beta_{12} + \beta_{11} N^*)^2} \left(\frac{\beta_1 N^*}{\beta_{11} + \beta_{12} N^*} - \pi_1 \alpha_1 \right) \left(\delta_1 F^* + \frac{rH^*}{L} \right), \\ C_4 &= \frac{k_1 A^* F^{*2} H^*}{(m + A^*)^2} \left(\alpha_0 + \frac{\beta_1 \beta_{12} A^*}{(\beta_{12} + \beta_{11} N^*)^2} \right) \left(\lambda_2 k_2^2 - \frac{r\delta_1}{L} \right) - \frac{\theta_1 \beta_1 \beta_{12} A^* F^* H^*}{(\beta_{12} + \beta_{11} N^*)^2} \left(\frac{\beta_1 N^*}{(\beta_{12} + \beta_{11} N^*)} - \pi_1 \alpha_1 \right) \left(\lambda_2 k_2^2 - \frac{r\delta_1}{L} \right) \\ &\quad + \frac{\lambda_1 k_1 m F^* H^*}{(m + A^*)^2} \left(\frac{k_1 r A^*}{L(m + A^*)} - \frac{pk_2 \lambda_2 \theta_1 \beta_1 \beta_{12} A^*}{(\beta_{12} + \beta_{11} N^*)^2} \right). \end{aligned}$$

Now, using Routh–Hurwitz criterion [30], we can say that roots of Eq. (25) lie in the left half of the complex plane, if and only if $C_j > 0$ for $(j = 1, 3, 4)$ and $C_3(C_1 C_2 - C_3) - C_1^2 C_4 > 0$. Thus, equilibrium E^* is locally asymptotically stable if $C_j > 0$ for $(j = 1, 3, 4)$ and $C_3(C_1 C_2 - C_3) - C_1^2 C_4 > 0$. $\square$

3.3 Bifurcation analysis

In this section, we investigate the bifurcation phenomena associated with the proposed mathematical model (8). Bifurcation analysis plays a crucial role in understanding the qualitative changes in the system's dynamics as key parameters vary. Specifically, we explore how variations in critical system parameters influence equilibrium stability and lead to the emergence of new dynamical behaviors. To show the existence of different bifurcations, we select β_1 as a bifurcation parameter. By systematically varying β_1 in our analysis, we can comprehensively explore how changes in nutrient utilization efficiency propagate through the entire ecosystem, from basal nutrient dynamics to higher trophic levels.

3.3.1 Transcritical bifurcation

Through the stability analysis of the obtained equilibria, it is evident that Jacobian matrix J (evaluated at equilibrium E_6) exhibits one eigenvalue that can be positive, negative, or zero depending on the specific values of the model parameters provided the remaining three eigenvalues are negative. This suggests the possible occurrence of a transcritical bifurcation [31] around the equilibrium E_6 . Thus, we have the following theorem regarding transcritical bifurcation for model system (8) around the equilibrium point E_6 as given below.

Theorem 4 At $\beta_1 = \beta_{1c} = \left(\alpha_1 + \frac{k_1 F_6}{m}\right) \left(\frac{\beta_{12}\alpha_0 + \beta_{11}(q+pH_6)}{\theta_1(q+pH_6)}\right)$, the direction of transcritical bifurcation of model system (8) around equilibrium E_6 is forward if $a < 0$ and backward if $a > 0$, where

$$a = 2 \left[\frac{\theta_1 \beta_{1c} \beta_{12}}{(\beta_{12} + \beta_{11} N_6)^2} \left(-\frac{\beta_{1c}(q + pH_6)}{\beta_{12}\alpha_0 + \beta_{11}(q + pH_6)} + \pi_1 \alpha_1 \right) + \frac{rk_1^2}{m^2} \left(\frac{\delta_0 - k_2 L + \lambda_1}{r\delta_1 + \lambda_2 k_2^2 L} \right) \right].$$

Proof To investigate this, we designate β_1 as the bifurcation parameter. Suppose there exists a $\beta_1 = \beta_{1c}$, such that eigenvalue $\psi_{62} = 0$, which gives $\beta_{1c} = \left(\alpha_1 + \frac{k_1 F_6}{m}\right) \left(\frac{\beta_{12}\alpha_0 + \beta_{11}(q+pH_6)}{\theta_1(q+pH_6)}\right)$. Thus, we apply the center manifold theorem [31] at E_6 . We first introduce $N = x_1$, $A = x_2$, $F = x_3$ and $H = x_4$, thus model system (8) can be written as

$$\begin{aligned} \frac{dx_1}{dt} &= q + px_4 - \alpha_0 x_1 - \beta_1 \frac{x_1 x_2}{\beta_{12} + \beta_{11} x_1} + \pi_1 \alpha_1 x_2 := g_1, \\ \frac{dx_2}{dt} &= \theta_1 \beta_1 \frac{x_1 x_2}{\beta_{12} + \beta_{11} x_1} - \alpha_1 x_2 - \frac{k_1 x_2 x_3}{m + x_2} := g_2, \\ \frac{dx_3}{dt} &= \left(\delta_0 - \frac{\lambda_1 k_1 x_2}{m + x_2} \right) x_3 - \delta_1 x_3^2 - k_1 x_3 x_4 := g_3, \\ \frac{dx_4}{dt} &= rx_4 \left(1 - \frac{x_4}{L} \right) + \lambda_2 k_2 x_3 x_4 := g_4. \end{aligned} \quad (26)$$

The Jacobian matrix for system (8) around the equilibrium E_6 at $\beta_1 = \beta_{1c}$ is

$$J_6 \Big|_{\beta_1 = \beta_{1c}} = \begin{bmatrix} -\alpha_0 & b_{12} & 0 & p \\ 0 & 0 & 0 & 0 \\ 0 & -b_{32} & -\delta_1 F_6 & -k_2 F_6 \\ 0 & 0 & b_{43} & -\frac{rH_6}{L} \end{bmatrix},$$

where

$$b_{12} = -\frac{\beta_{1c}(q + pH_6)}{\beta_{12}\alpha_0 + \beta_{11}(q + pH_6)} + \pi_1 \alpha_1, \quad b_{32} = \frac{\lambda_1 k_1 F_6}{m}, \quad b_{43} = \lambda_2 k_2 H_6.$$

Thus, the right eigenvector ($\mathfrak{U}$) and left eigenvector ($\mathfrak{W}$) associated with eigenvalue 0 are represented as follows

$$\mathfrak{U} = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{bmatrix} = \begin{bmatrix} \frac{1}{\alpha_0} \left(b_{12} - \frac{\lambda_1 \lambda_2 k_1 k_2 p L}{m(\delta_1 r + \lambda_2 k_2^2 L)} \right) \\ 1 \\ -\frac{\lambda_1 k_1 r}{m(\delta_1 r + \lambda_2 k_2^2 L)} \\ -\frac{\lambda_1 \lambda_2 k_1 k_2^2 L}{m(\delta_1 r + \lambda_2 k_2^2 L)} \end{bmatrix} \text{ and } \mathfrak{W} = \begin{bmatrix} w_1 \\ w_2 \\ w_3 \\ w_4 \end{bmatrix}^T = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}^T.$$

The coefficients a and b of Theorem 4.1 in [31] for system (26) are written as

$$a = \sum_{i,j,k=1}^4 w_k u_i u_j \frac{\partial^2 g_k}{\partial x_i \partial x_j}(E_6, \beta_{1c}) \text{ and } b = \sum_{i,k=1}^4 w_k u_i \frac{\partial^2 g_k}{\partial x_i \partial \beta_1}(E_6, \beta_{1c}).$$

Here, we obtain the value of a and b for model system (26) as follows

$$a = 2 \left[\frac{\theta_1 \beta_{1c} \beta_{12}}{(\beta_{12} + \beta_{11} N_6)^2} \left(-\frac{\beta_{1c}(q + p H_6)}{\beta_{12} \alpha_0 + \beta_{11}(q + p H_6)} + \pi_1 \alpha_1 \right) + \frac{r k_1^2}{m^2} \left(\frac{\delta_0 - k_2 L + k_1 \lambda_1}{r \delta_1 + \lambda_2 k_2^2 L} \right) \right],$$

$$b = \frac{\theta_1 (q + p H_6)}{\beta_{12} \alpha_0 + \beta_{11}(q + p H_6)}.$$

Here, $b > 0$, and a can be either positive or negative, depending on the parameter values. Thus, if $a > 0$, the transcritical bifurcation occurs in the backward direction, whereas if $a < 0$, the bifurcation proceeds in the forward direction. $\square$

Remark 2 The sign of the coefficient a determines the direction of the transcritical bifurcation for model system (8), which has critical implications for the stability and existence of equilibria. When $a > 0$, the transcritical bifurcation occurs in the backward direction, meaning that an interior equilibrium becomes stable as the bifurcation parameter decreases below a critical threshold. In this scenario, the system exhibits hysteresis, where nutrient, algae, fish population and human coexist, even under the unfavorable conditions until a tipping point is reached. Conversely, when $a < 0$, the bifurcation proceeds in the forward direction, indicating that the interior equilibrium gains stability as the bifurcation parameter increases above the critical value, leading to a smooth transition between stable and unstable states. The direction of bifurcation thus hinges on the delicate balance between parameters related nutrient availability, algal toxicity and human harvesting pressure on fish population. This analysis reveals how shifts in environmental or anthropogenic factors can abruptly alter system's stability.

3.3.2 Saddle-node bifurcation

From the equilibrium analysis, it is clear that the model system (8) can exhibit two or three interior equilibrium depending on the parameter values. Thus, there may be a chance that model system (8) can exhibit saddle-node bifurcation [32, 33]. Therefore, in this section, we explore the condition under which model system (8) undergoes saddle-node bifurcation.

Theorem 5 For the existence of saddle-node bifurcation around the interior equilibrium E^* , there exists a $\beta_1 = \beta_{1r}$, such that $\mathfrak{B}_2 \neq 0$.

Proof We assume that $C_4(\beta_1 = \beta_{1r}) = 0$. Thus, the Jacobian matrix of model system (8) around interior equilibrium E^* at $\beta_1 = \beta_{1r}$ is

$$J^*|_{\beta_1=\beta_{1r}} = \begin{bmatrix} -\tilde{b}_{11} & -\tilde{b}_{12} & 0 & p \\ \tilde{b}_{21} & \tilde{b}_{22} & -\tilde{b}_{23} & 0 \\ 0 & -\tilde{b}_{32} & -\delta_1 F^* & -k_2 F^* \\ 0 & 0 & \lambda_2 k_2 H^* & -\frac{r H^*}{L} \end{bmatrix},$$

where

$$\tilde{b}_{11} = \alpha_0 + \frac{\beta_{1r} \beta_{12} A^*}{(\beta_{12} + \beta_{11} N^*)^2}, \quad \tilde{b}_{12} = \frac{\beta_{1r} N^*}{\beta_{12} + \beta_{11} N^*} - \pi_1 \alpha_1, \quad \tilde{b}_{21} = \frac{\theta_1 \beta_{1r} \beta_{12} A^*}{(\beta_{12} + \beta_{11} N^*)^2},$$

$$\tilde{b}_{22} = \frac{k_1 A^* F^*}{(m + A^*)^2}, \quad \tilde{b}_{23} = \frac{k_1 A^*}{m + A^*}, \quad \tilde{b}_{32} = \frac{\lambda_1 k_1 m F^*}{(m + A^*)^2}.$$

The Jacobian matrix $J^*|_{\beta_1=\beta_{1r}}$ has one simple 0 eigenvalue. Let $\tilde{\mathcal{U}} = [\tilde{u}_1 \ \tilde{u}_2 \ \tilde{u}_3 \ \tilde{u}_4]^\top$ and $\tilde{\mathcal{W}} = [\tilde{w}_1 \ \tilde{w}_2 \ \tilde{w}_3 \ \tilde{w}_4]$ are sequentially represent the right and left eigenvectors of matrix J^* corresponding to the 0 eigenvalue, where

$$\tilde{u}_1 = \tilde{b}_{12} F^* (\delta_1 F^* + \lambda_2 k_2^2 L) + \lambda_2 k_2 p L \tilde{b}_{32}, \quad \tilde{u}_2 = -\tilde{b}_{11} F^* (\delta_1 F^* + \lambda_2 k_2^2 L), \quad \tilde{u}_3 = \lambda_2 k_2 L \tilde{b}_{11} \tilde{b}_{32}, \quad \tilde{u}_4 = r \tilde{b}_{11} \tilde{b}_{32},$$

$$\tilde{w}_1 = \tilde{b}_{21} \tilde{b}_{32}, \quad \tilde{w}_2 = \tilde{b}_{11} \tilde{b}_{32}, \quad \tilde{w}_3 = -\tilde{b}_{12} \tilde{b}_{21} + \tilde{b}_{11} \tilde{b}_{22}, \quad \tilde{w}_4 = \frac{L}{r H^*} (p \tilde{b}_{21} - k_2 \tilde{b}_{11} F^*) \tilde{b}_{32}.$$

Suppose, $\mathcal{F} = (f_1, f_2, f_3, f_4)^\top$, where f_1, f_2, f_3 , and f_4 are sequentially the right hand side of $dN/dt, dA/dt, dF/dt$ and dH/dt of model system (8). Thus,

$$\mathfrak{B}_1 = \tilde{\mathcal{W}} \cdot \frac{\partial \mathcal{F}}{\partial \beta_1} \Big|_{(E^*, \beta_1=\beta_{1r})} = \frac{\alpha_0 \theta_1 \lambda_1 k_1 m N^* A^* F^*}{(m + A)^2 (\beta_{12} + \beta_{11} N^*)} > 0,$$

$$\mathfrak{B}_2 = \mathcal{W} [\mathcal{C}^2 \mathcal{Q}(\mathcal{U}, \mathcal{U})] \Big|_{(E^*, \beta_1=\beta_{1r})}$$

$$= 2 \left[\frac{\beta_{1r} \beta_{11} \beta_{12} A^*}{(\beta_{12} + \beta_{11} N^*)^3} \tilde{u}_1 - \frac{\beta_{1r} \beta_{12}}{(\beta_{12} + \beta_{11} N^*)^2} \tilde{u}_2 \right] \tilde{u}_1 \tilde{w}_1 + 2 \left[\frac{\theta_1 \beta_{1r} \beta_{11} \beta_{12} A^*}{(\beta_{12} + \beta_{11} N^*)^3} \tilde{u}_1^2 - \frac{k_1 A^* F^*}{(m + A^*)^3} \tilde{u}_2^2 + \frac{k_1 A^*}{(m + A^*)^2} \tilde{u}_2 \tilde{u}_3 \right] \tilde{w}_2$$

$$+ \frac{2\lambda_1 k_1 m F^*}{(m + A^*)^3} \tilde{u}_2^2 \tilde{w}_3.$$

Thus, according to the Sotomayor's theorem, the conditions required for the existence of a saddle-node bifurcation are satisfied when $\mathfrak{B}_2 \neq 0$. $\square$

Remark 3 The condition $\mathfrak{B}_2 \neq 0$ captures the nonlinear interactions in the system near the equilibrium point E^* and ensures that the bifurcation is non-degenerate. Also, $\mathfrak{B}_2 \neq 0$ confirms that small parameter changes in bifurcation parameter β_1 can lead to abrupt ecological shifts, such as the collapse or emergence of stable equilibrium, highlighting the system's sensitivity to critical thresholds in nutrient availability or species interactions.

3.3.3 Hopf bifurcation

To show the existence of Hopf bifurcation [32] around coexisting equilibrium E^* , we again select β_1 as a bifurcation. Therefore, we have the following theorem.

Theorem 6 For Hopf bifurcation to occur around the interior equilibrium E^* , the necessary and sufficient conditions are that there exists a critical value of $\beta_1 = \beta_{1e}$, such that

- (i) $C_i(\beta_{1e}) > 0$ for $i = 3, 4$,
- (ii) $C_3(\beta_{1e})(C_1(\beta_{1e})C_2(\beta_{1e}) - C_3(\beta_{1e})) - C_1^2(\beta_{1e})C_4(\beta_{1e}) = 0$,
- (iii) $\left. \frac{d}{d\beta_1}(C_1C_2C_3 - C_3^2 - C_1^2C_4) \right|_{\beta_1=\beta_{1e}} \neq 0$.

Proof Consequently, we express each coefficient of the characteristic Eq. (25) (i.e., C_i) as a function of β_1 .

$$\Psi^4 + C_1(\beta_1)\Psi^3 + C_2(\beta_1)\Psi^2 + C_3(\beta_1)\Psi + C_4(\beta_1) = 0. \quad (27)$$

Here, we assume that there exists a $\beta_1 = \beta_{1e}$, such that $C_i(\beta_{1e}) > 0$ for $i = 3, 4$ and $C_3(\beta_{1e})(C_1(\beta_{1e})C_2(\beta_{1e}) - C_3(\beta_{1e})) - C_1^2(\beta_{1e})C_4(\beta_{1e}) = 0$. Thus, Eq. (27) can be written as

$$\left(\Psi^2 + \frac{C_3}{C_1} \right) + \left(\Psi^2 + C_1\Psi + \frac{C_1C_4}{C_3} \right), \quad (28)$$

which gives four roots Ψ_i ($i = 1, 2, 3, 4$) with a pair of purely imaginary roots $\Psi_{1,2} = \pm i\omega_0$, where $\omega_0 = \sqrt{\frac{C_3}{C_1}}$. The nature of rest of the roots $\Psi_{3,4}$ can be identified with the following set of equations

$$\begin{aligned} \Psi_3 + \Psi_4 &= -C_1, \\ \omega_0^2 + \Psi_3\Psi_4 &= C_2, \\ \omega_0^2(\Psi_3 + \Psi_4) &= -C_3, \\ \omega_0^2\Psi_3\Psi_4 &= C_4. \end{aligned} \quad (29)$$

Thus, from Eq. (29), we can say that the roots Ψ_3 and Ψ_4 always lie in the left half of the complex plane. Now, we select a point in the ϵ -neighborhood of β_{1p} , such that $\Psi_{1,2} = \omega(\beta_1) \pm i\xi(\beta_1)$. After substituting $\Psi_{1,2} = \omega(\beta_1) \pm i\xi(\beta_1)$ into Eq. (27) and collecting the imaginary and real parts, we obtain

$$\omega^4 + C_1\omega^3 + C_2\omega^2 + C_3\omega + C_4 + \xi^4 - 6\omega^2\xi^2 - 3C_1\omega\xi^2 - C_2\xi^2 = 0, \quad (30)$$

and

$$4\omega\xi(\omega^2 - \xi^2) - C_1\xi^3 + 3C_1\omega^2\xi + 2C_2\omega\xi + C_3\xi = 0. \quad (31)$$

As $\xi(\beta_1) \neq 0$, thus from Eq. (31), we have $4\omega(\omega^2 - \xi^2) - C_1\xi^2 + 3C_1\omega^2 + 2C_2\omega + C_3 = 0$. Now substituting it into Eq. (30) and differentiating with respect to β_1 , we get

$$\left. \frac{d\omega}{d\beta_1} \right|_{\beta_1=\beta_{1e}} = \left[\frac{\frac{d}{d\beta_1}(C_1C_2C_3 - C_3^2 - C_1^2C_4)}{-2C_1(C_1C_3 + (2C_3/C_1 - C_2)^2)} \right]_{\beta_1=\beta_{1e}} \neq 0, \text{ provided } \frac{d}{d\beta_1}(C_1C_2C_3 - C_3^2 - C_1^2C_4) \neq 0. \quad (32)$$

Thus, we can say that the transversality condition holds if $\left. \frac{d}{d\beta_1}(C_1C_2C_3 - C_3^2 - C_1^2C_4) \right|_{\beta_1=\beta_{1e}} \neq 0$. $\square$

Table 3 Parameter values for model system (8)

Parameter	Value	Parameter	Value	Value	Parameter
q	4	p	0.01	α_0	0.006
β_1	0.55	β_{12}	0.5	β_{11}	1
π_1	0.3	α_1	0.5	θ_1	1.2
m	5	λ_1	0.5	λ_2	0.05
k_1	0.01	k_2	0.005	r	0.01
δ_0	0.5	L	50	δ_1	0.03

Remark 4 The conditions for Hopf bifurcation around the interior equilibrium E^* reveal fundamental insights about the emergence of oscillatory dynamics in the nutrient–algae–fish–human ecosystem. The three key mathematical requirements—(i) $C_i(\beta_{1e}) > 0$ for $i = 3, 4$, (ii) $C_3(\beta_{1e})(C_1(\beta_{1e})C_2(\beta_{1e}) - C_3(\beta_{1e})) - C_1^2(\beta_{1e})C_4(\beta_{1e}) = 0$, and (iii) $\left. \frac{d}{d\beta_1}(C_1C_2C_3 - C_3^2 - C_1^2C_4) \right|_{\beta_1=\beta_{1e}} \neq 0$, collectively ensure that as the nutrient uptake rate β_1 crosses its critical value β_{1e} , the system undergoes a qualitative transition from stable equilibrium to periodic oscillations. Biologically, this bifurcation corresponds to the onset of cyclical population dynamics, where nutrient concentrations, algal blooms, fish populations, and human impacts begin to oscillate in a sustained manner.

4 Numerical simulations

In this section, we perform numerical simulations for the model system (8) to validate the analytical results derived in the previous section. Due to the limited availability of real-world data related to our model, we have chosen a representative set of parameter values for the simulations, i.e., represented in Table 3. The numerical simulations aim to confirm the theoretical findings and provide deeper insights into the dynamics of the system under study. For the set of parameter values mentioned in Table 3, the components of interior equilibrium $E^*(N^*, A^*, F^*, H^*)$ are

$$N^* \approx 1.14, \quad A^* \approx 66.70, \quad F^* \approx 6.77, \quad H^* \approx 58.46.$$

Additionally, the eigenvalues of Jacobian matrix J^* are

$$-0.014, \quad -0.20, \quad -0.53 + 0.24i, \quad -0.53 - 0.24i,$$

that are either negative or contain negative real parts, which confirms the local stability of interior equilibrium E^* .

4.1 Effect of some parameters on dynamical variables

To explore the impact of HABs on fish, we plot the variation diagram by varying the grazing rate parameter k_1 from 0.05 to 0.1 (rest of the parameter values are same as mentioned in Table 3), Fig. 2. Through this figure, we examine how changes in k_1 influence the equilibrium levels of nutrient concentration, algae concentration, fish population, and human population. From Fig. 2b, it is observed that as the grazing rate k_1 increases, the fish population exerts stronger predation pressure on HABs, leading to a significant reduction in algae density at equilibrium. However, this reduction in algae biomass alters the nutrient dynamics, resulting in an increase in the nutrient concentration, Fig. 2a. This outcome occurs because algae plays a crucial role in nutrient uptake and assimilation.

Although an increase in the grazing rate k_1 leads to a reduction in algae biomass, the equilibrium levels of both the fish population and the human population decline as k_1 increases. This counterintuitive result is visually represented in Fig. 2c–d. The decline in the fish population can be attributed to the harmful effects of algae, which persist even as their concentration decreases. Algae, particularly in the form of HABs, can produce toxins, deplete oxygen levels, and disrupt the aquatic food web, all of which negatively impact fish health and survival. As a result, the equilibrium size of the fish population diminishes. Moreover, the reduction in the human population linked to the interconnectedness of the ecological system. A decrease in fish abundance may directly affect human activities, such as fishing, which depend on fish as a resource. These findings highlight the complex trade-offs associated with using fish as a biological control mechanism for HABs. While increased grazing can effectively mitigate algal blooms and nutrient concentrations, it may also have adverse effects on fish population and the human communities that rely on them. This underscores the need for a balanced approach when designing strategies to manage HABs, taking into account both ecological dynamics and socioeconomic considerations.

To investigate the impact of the parameter β_1 , which governs the rate of nutrient uptake by algae, we draw variation plot by varying the value of β_1 from 0.3 to 0.5 (with $q = 1$, $p = 0.1$, $\alpha_0 = 0.6$, $k_1 = .01$, $\beta_{12} = 0.03$, $\pi_1 = 0.03$ and rest of the parameter values are the same as mentioned in Table 3), Fig. 3. This figure allows us to observe how changes in the efficiency of nutrient uptake by algae influence the equilibrium levels of system variables. From Fig. 3a, it is clear that as β_1 increases, the equilibrium level of nutrient concentration decreases. This is because a higher value of β_1 represents more efficient nutrient uptake by algae, leading to

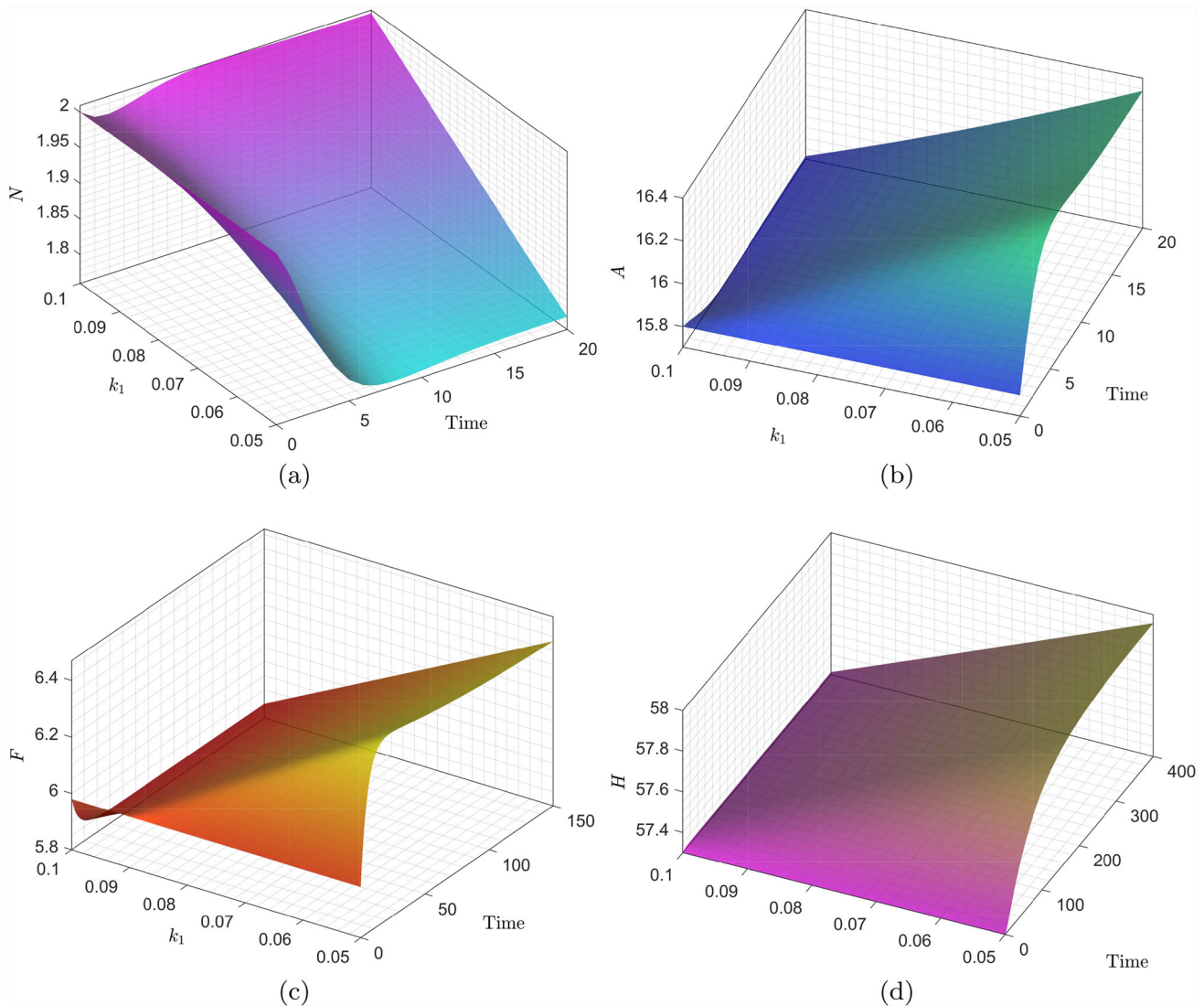


Fig. 2 Surface plot by varying the value of k_1 from 0.05 to 0.1 for (a) nutrient density (b) algae density (c) fish population (d) human population. Here, all the parameter values are the same as mentioned in Table 3 except k_1

a faster depletion of available nutrients in the system. However, this enhanced nutrient uptake also stimulates the subsequent growth of HABs, as shown in Fig. 3b. The effect of β_1 on the fish population is shown in Fig. 3c. From this figure, we can see that for moderate values of β_1 , there is little impact on the fish population. However, once β_1 exceeds a certain threshold, the fish population experiences a significant decline. This decline is likely due to the increase in HABs biomass, which can lead to detrimental effects such as toxin production and habitat degradation, ultimately reducing the survivability of fish in the ecosystem. Similarly, the human population follows a trend similar to that of the fish population, as depicted in Fig. 3d. Initially, moderate increases in β_1 have minimal effects on the human population. However, once β_1 surpasses a critical value, the increase in HABs begins to significantly reduce the human population.

4.2 Bifurcation results

We extend our analysis by numerically investigating the presence of key bifurcations, including transcritical, saddle-node, and Hopf bifurcations. To explore these phenomena, we construct bifurcation diagrams for various scenarios. Specifically, Figs. 4, 5, 6, 7, 8 present bifurcation diagrams generated for different values of the parameter k_1 , while treating β_1 as a bifurcation parameter. These diagrams allow us to identify critical transitions in the system's dynamics, providing insights into the conditions under which qualitative changes, such as the emergence or loss of stability, occur. For the bifurcation results, we set $p = 0.1$, $\alpha_0 = 0.066$, $\pi_1 = 0.03$, $\lambda_1 = 0.3$ and rest of the parameter values are the same as mentioned in Table 3.

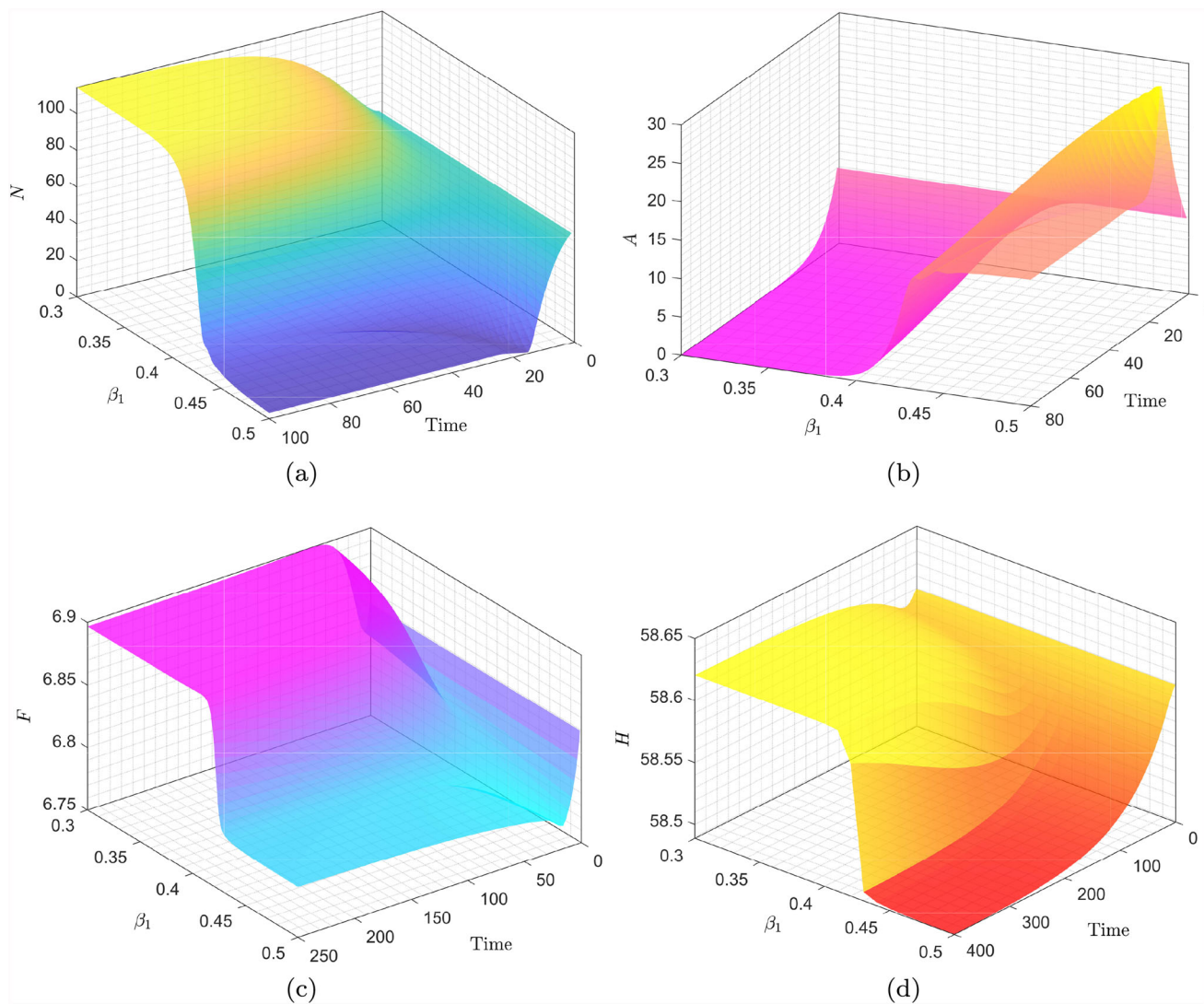


Fig. 3 Surface plot by varying the value of β_1 from 0.3 to 0.5 for (a) nutrient density (b) algae density (c) fish population (d) human population. Here, $q = 1$, $p = 0.1$, $\alpha_0 = 0.6$, $k_1 = .01$, $We\beta_{12} = 0.03$, $\pi_1 = 0.03$ and rest of the parameter values are the same as mentioned in Table 3

Figure 4a illustrates the bifurcation diagram for $k_1 = 0.0001$, reveals a clear existence of a forward transcritical bifurcation for model system (8). A forward transcritical bifurcation occurs when the stability of equilibrium points changes as a system parameter is varied. In Fig. 4a, the transcritical bifurcation point is denoted as BP (at $\beta_1 \approx 0.418$). This point represents a critical threshold beyond which the system experiences a dramatic shift in the stability of its equilibrium points. Before reaching the bifurcation point BP, the system exhibits a stable boundary equilibrium E_6 . In this regime, the model describes a scenario, where HABs are unable to grow significantly due to the insufficient nutrient availability. As a result, the algae-free equilibrium is stable, and the concentration of HABs remains low. However, once the value of the parameter β_1 surpasses the critical threshold at point BP, the system transitions to a new regime, where the interior equilibrium point E^* , representing a viable HAB biomass, becomes stable. In this regime, the HAB biomass experiences significant growth due to the increased availability of nutrients, rendering the interior equilibrium point stable. This shift in stability signifies a fundamental change in the dynamics of the ecosystem. The equilibrium E_6 , which was previously stable at lower values of β_1 , becomes unstable as the HAB concentration uptake nutrients rapidly. The nutrient dynamics, which initially restricted HAB growth, now support the proliferation of algae, leading to a new stable state with a higher HAB concentration.

To further analyze the behavior of the model system (8) around the bifurcation point BP, we have generated variation plots for $\beta_1 = 0.415$ and $\beta_1 = 0.4186$, as shown in Fig. 4b. These plots provide a detailed analysis of the system's dynamics both before and after the bifurcation point BP. From Fig. 4b, it is evident that for $\beta_1 = 0.415$, which is below the bifurcation threshold, all solution trajectories gravitate toward the boundary equilibrium. This indicates that under these conditions, the system dynamics are dominated by the algae-free equilibrium, and the HAB biomass fails to establish significant growth. The system remains in a

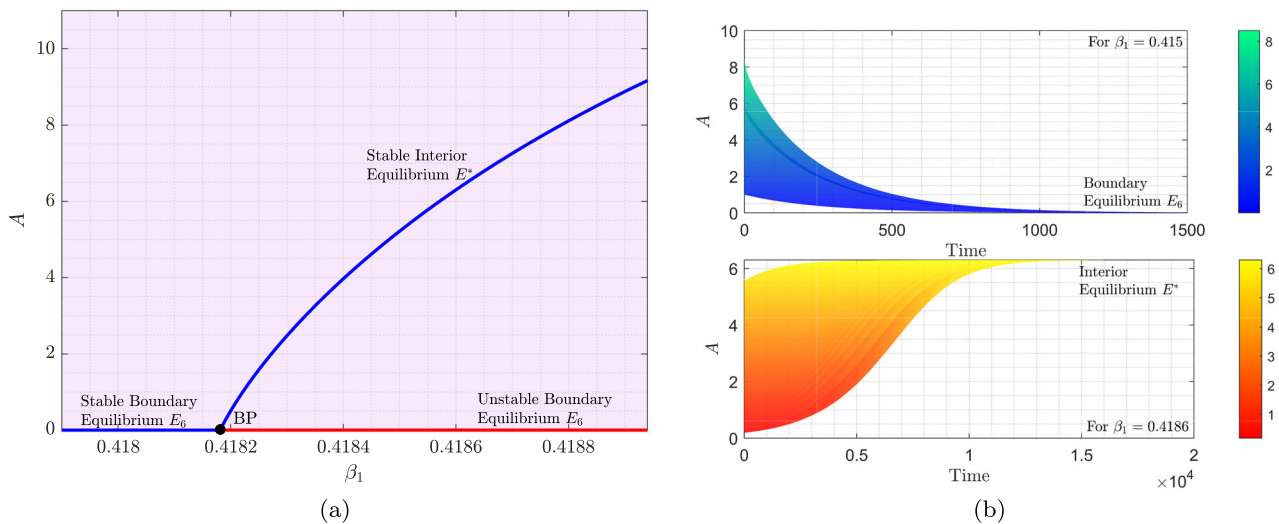


Fig. 4 **a** Showing the occurrence of transcritical bifurcation in forward direction at $\beta_1 \approx 0.418$ **(b)** variation plot for $\beta_1 = 0.415$ and $\beta_1 = 0.4186$. Here, $k_1 = 0.001$, $p = 0.1$, $\alpha_0 = 0.066$, $\pi_1 = 0.03$, $\lambda_1 = 0.3$ and rest of the parameter values are the same as mentioned in Table 3

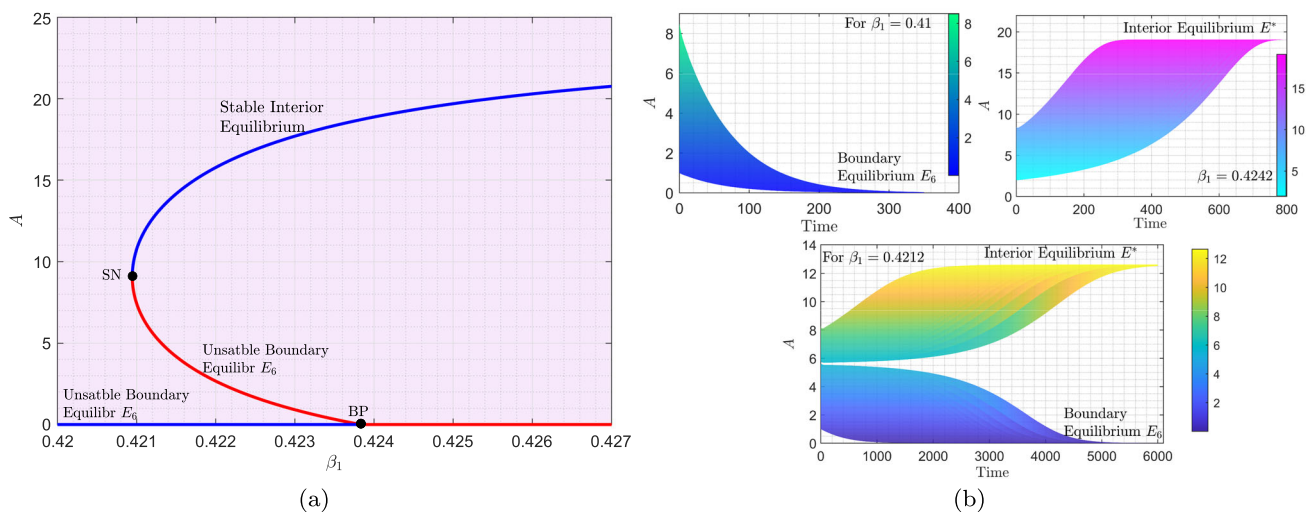


Fig. 5 **a** Showing the occurrence of transcritical bifurcation in backward direction at $\beta_1 \approx 0.424$ and saddle-node bifurcation at $\beta_1 \approx 0.421$ **(b)** variation plot for $\beta_1 = 0.41$, $\beta_1 = 0.4242$ and $\beta_1 = 0.4212$. Here, $k_1 = 0.005$, $p = 0.1$, $\alpha_0 = 0.066$, $\pi_1 = 0.03$, $\lambda_1 = 0.3$ and rest of the parameter values are the same as mentioned in Table 3

state where nutrient uptake and other limiting factors suppress the proliferation of HABs. In contrast, for $\beta_1 = 0.4186$, which is slightly above the bifurcation threshold, the solution trajectories exhibit a markedly different behavior. In this regime, all trajectories gravitate toward the interior equilibrium, reflecting a shift in the system dynamics. The increase in β_1 facilitates the growth of HABs, enabling the system to stabilize around an interior equilibrium that supports a viable biomass of HABs. The exchange of stability between equilibria at BP point mirrors real-world observations where small changes in nutrient loading can trigger irreversible algal blooms. These observations underscore the critical influence of the bifurcation point on the system's dynamics. The transition from boundary equilibrium to interior equilibrium as β_1 crosses the threshold highlights the sensitivity of the model to parameter variations and provides valuable insights into the conditions that enable or suppress HAB growth within the aquatic ecosystem.

The bifurcation diagram presented in Fig. 5a for the parameter value $k_1 = 0.005$ reveals a complex dynamical landscape for the model system (8), characterized by two distinct bifurcation points. At $\beta_1 \approx 0.4209$, the system undergoes a saddle-node bifurcation (denoted SN), where two interior equilibria coalesce and disappear, while at $\beta_1 \approx 0.4238$, a transcritical bifurcation (denoted BP) occurs, marking an exchange of stability between equilibrium points. These two critical points partition the system's behavior into three biologically significant regimes that reflect different ecological states of the aquatic ecosystem. For $\beta_1 < 0.4209$, the complete absence of interior equilibria indicates an ecological state, where environmental conditions, particularly nutrient availability and growth factors, remain insufficient to sustain any meaningful HAB biomass. In this regime, all solution trajectories consistently converge to the boundary equilibrium (as clearly demonstrated in Fig. 5b for $\beta_1 = 0.41$), representing a stable, algae-free aquatic

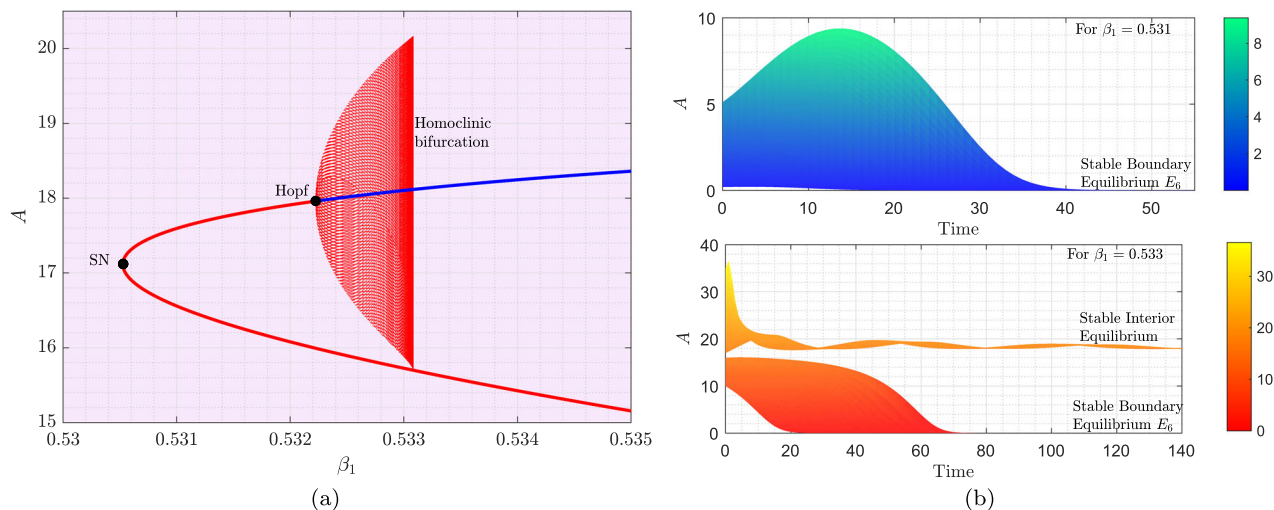


Fig. 6 **a** Showing the occurrence of subcritical Hopf bifurcation at $\beta_1 \approx 0.532$ and saddle-node bifurcation at $\beta_1 \approx 0.530$ **(b)** variation plot for $\beta_1 = 0.531$ and $\beta_1 = 0.533$. Here, $k_1 = 0.4$, $p = 0.1$, $\alpha_0 = 0.066$, $\pi_1 = 0.03$, $\lambda_1 = 0.3$ and rest of the parameter values are the same as mentioned in Table 3

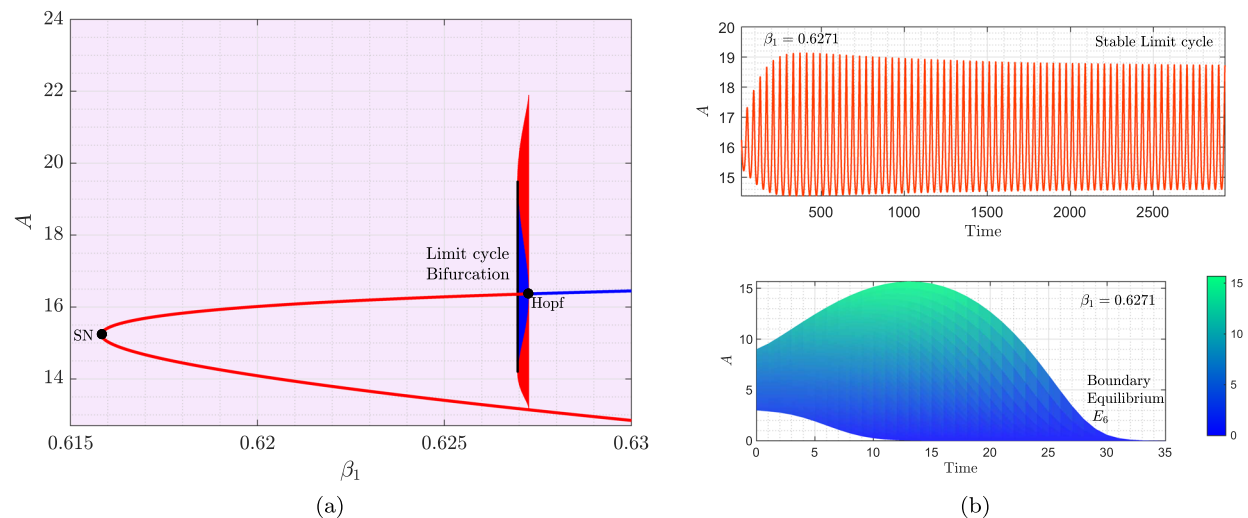


Fig. 7 **a** Showing the occurrence of supercritical Hopf bifurcation at $\beta_1 \approx 0.6272$, saddle-node bifurcation at $\beta_1 \approx 0.6158$ and limit cycle bifurcation at $\beta_1 \approx 0.6270$ **(b)** variation plot for $\beta_1 = 0.6271$. Here, $k_1 = 0.67$, $p = 0.1$, $\alpha_0 = 0.066$, $\pi_1 = 0.03$, $\lambda_1 = 0.3$ and rest of the parameter values are the same as mentioned in Table 3

environment that maintains ecological balance and supports diverse aquatic life. This regime corresponds to healthy water bodies, where nutrient concentrations are well below the threshold required for algal bloom formation. The intermediate range $\beta \in (0.4209, 0.4238)$ represents a critical transition zone, where the system exhibits bistability, with coexistence of two interior equilibria (shown for $\beta_1 = 0.4212$ in Fig. 5b). The lower equilibrium point, typically unstable, acts as an ecological threshold separating the basin of attraction for the boundary equilibrium from that of the higher, stable interior equilibrium. This regime reflects an ecologically vulnerable state, where minor perturbations in nutrient levels or environmental conditions can trigger a regime shift from an algae-free state to one dominated by HABs. The presence of alternative stable equilibria in this parameter range highlights the potential for hysteresis effects, where the system may resist returning to its original state even after nutrient reductions. When β_1 exceeds 0.4238, the system transitions to a monostable regime dominated by a single, stable interior equilibrium (shown for $\beta_1 = 0.4242$ in Fig. 5b), representing an ecosystem state characterized by persistent and potentially harmful algal blooms. In this regime, the high nutrient availability drives the system inexorably toward an algal-dominated state, regardless of initial conditions. This corresponds to eutrophic water bodies, where excessive nutrient loading has fundamentally alter the ecosystem's dynamics, leading to degraded water quality and reduced biodiversity. The trajectory behavior across these regimes, as illustrated in Fig. 5b, provides valuable insights for ecosystem management. The existence of multiple stable states in the intermediate regime suggests that both preventive

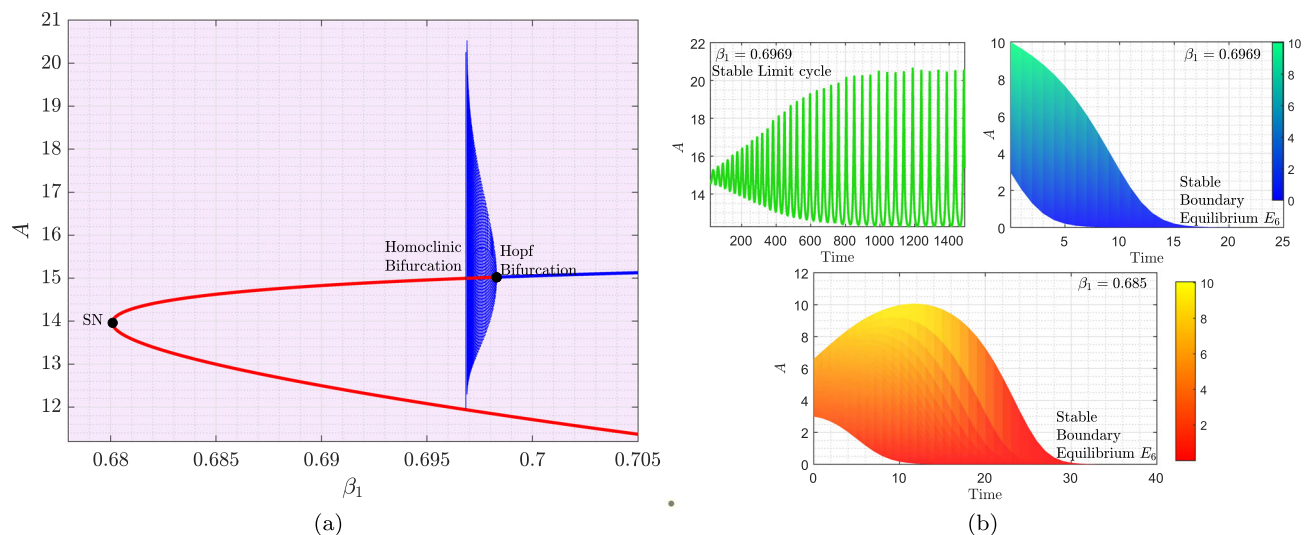


Fig. 8 **a** Showing the occurrence of supercritical Hopf bifurcation at $\beta_1 \approx 0.698$ and saddle-node bifurcation at $\beta_1 \approx 0.68$ **(b)** variation plot for $\beta_1 = 0.6969$ and $\beta_1 = 0.685$. Here, $k_1 = 0.85$, $p = 0.1$, $\alpha_0 = 0.066$, $\pi_1 = 0.03$, $\lambda_1 = 0.3$ and rest of the parameter values are the same as mentioned in Table 3

measures (maintaining β_1 below 0.4209) and active intervention strategies (reducing β_1 below critical thresholds following bloom formation) may be necessary for effective HAB control.

For $k_1 = 0.4$, the bifurcation plot shown in Fig. 6a reveals the significant dynamic transitions in the model system. Notably, the system undergoes a subcritical Hopf bifurcation at $\beta_1 \approx 0.532$. This bifurcation leads to the emergence of an unstable limit cycle around the interior equilibrium, which corresponds to the equilibrium branch with high density of HABs. The unstable limit cycle reflects periodic fluctuations in the system's state, with HABs biomass oscillating in a manner that indicates unstable dynamics. However, this limit cycle eventually diminishes due to the occurrence of a homoclinic bifurcation, where the system's trajectories return to the same point, signaling a transition to a different stable regime. In addition to the Hopf bifurcation, the model system (8) also exhibits a saddle-node bifurcation at $\beta_1 \approx 0.530$. We have also generated variation plots for the parameter values $\beta_1 = 0.531$ and $\beta_1 = 0.533$, as shown in Fig. 6b. These plots illustrate how the trajectories of the system evolve in regions, where an unstable limit cycle exists. From Fig. 6b, we observe distinct behaviors in the system dynamics for the chosen values of β_1 . Specifically, when $\beta_1 = 0.531$, all solution trajectories converge toward the interior equilibrium point E_6 . This indicates that the system stabilizes at this equilibrium state, where no HAB biomass is present in the water body, regardless of the initial conditions of the dynamical variables. However, for $\beta_1 = 0.533$, the behavior of the trajectories becomes more complex. In this case, the solution trajectories either converge toward the stable interior equilibrium point or approach a boundary equilibrium point. The eventual outcome depends on the initial conditions of the system's dynamical variables.

We have also generated a bifurcation plot for the parameter value $k_1 = 0.67$, Fig. 7a. This figure illustrates that for $k_1 = 0.67$, the model system (8) undergoes a limit cycle bifurcation at 0.6270, via supercritical Hopf bifurcation that occurs at $\beta_1 \approx 0.6272$, corresponding to an equilibrium state characterized by a high algae density. Following this bifurcation, the system exhibits two distinct limit cycles: an inner limit cycle, which is stable, and an outer limit cycle, which is unstable. To further explore the dynamics of the system for the range of β_1 values where these two limit cycles coexist, we have generated variation plots for $\beta_1 = 0.6271$, Fig. 7b. This figure shows that the solution trajectories exhibit two possible outcomes: they either converge to the stable limit cycle or gravitate toward the boundary equilibrium point E_6 . These results have significant ecological implications, showing how aquatic systems can exhibit both stable periodic blooms and sudden collapse to algae-free states within narrow parameter ranges. The coexistence of stable and unstable limit cycles particularly highlights the system's sensitivity to initial conditions and the potential for hysteresis effects in bloom formation.

We have further generated a bifurcation plot for $k_1 = 0.85$, presented in Fig. 8a. This figure demonstrates that the model system (8) undergoes a supercritical Hopf bifurcation at $\beta_1 \approx 0.698$. As a result of this bifurcation, a stable limit cycle emerges from the interior equilibrium point associated with equilibrium of HABs biomass. However, this stable limit cycle eventually disappears due to the occurrence of a homoclinic bifurcation. To examine the behavior of solution trajectories within the range of β_1 values, where the stable limit cycle exists, we have generated variation plots, Fig. 8b. From this figure, it is evident that for $\beta_1 = 0.6969$, all solution trajectories exhibit one of two behaviors: they either converge toward the stable limit cycle or gravitate toward the stable boundary equilibrium depending on the initial condition. Additionally, Fig. 8b reveals the dynamics for $\beta_1 = 0.685$, a value within the range, where two unstable interior equilibria exist. For this parameter value, all solution trajectories eventually gravitate toward the stable boundary equilibrium.

5 Results and discussion

The results presented in this research work provide a comprehensive analysis of the intricate dynamics involved in the interactions between HABs, nutrients, aquatic fauna, and human population. Through the formulation of a mathematical model, we have elucidated the underlying mechanisms that govern these interactions and their implications for both ecological and human health. The results demonstrate that excessive nutrient levels create favorable conditions for algae growth, leading to the formation of HABs that can severely disrupt aquatic ecosystems. This not only threatens the biodiversity of aquatic life but also poses risks to human health, particularly through the contamination of drinking water sources and seafood. The model highlights the delicate balance that exists within aquatic ecosystems, where changes to one component can have cascading effects throughout the system.

Algal blooms, particularly harmful algal blooms, play a dual role in aquatic ecosystems, serving as both a primary food source for fish and a potential threat when they become toxic. While fish populations typically rely on algae as a vital food source, the presence of toxic algal blooms alters this dynamics, leading to decreased fish populations and, consequently, impacting fishery-dependent human communities. The study also emphasizes the importance of understanding the potential socioeconomic consequences of HABs. The decline in fish population due to algae toxicity can lead to substantial economic losses for fishing communities and affect food security. Moreover, the adverse effects on tourism and recreational activities in affected areas further amplify the economic burden associated with HABs. Therefore, effective management of nutrient run-off and proactive measures to monitor and mitigate algae blooms is important for preserving both ecological integrity and human livelihoods.

Moreover, the proposed model demonstrates the existence of transcritical bifurcation at a critical value of the nutrient uptake rate by HABs. When the nutrient uptake rate by HABs exceeds this critical threshold, the system undergoes a qualitative change, i.e., the interior equilibrium (where HABs coexist with fish and humans) becomes feasible, while the HABs-free equilibrium (where HABs are absent) loses its stability. The underlying mechanism for this phenomenon is that when the nutrient uptake rate by HABs is relatively low, the density of HABs remains insufficient to sustain their population within the water body. As a result, the system stabilizes at a HABs-free equilibrium, where HABs are effectively absent. However, as the nutrient uptake rate increases beyond the critical threshold, HABs gain access to sufficient nutrients to support their growth and proliferation. This leads to a shift in the system's dynamics, where the interior equilibrium characterized by the coexistence of HABs with other components such as nutrients, fish, and humans becomes feasible and stable. At the same time, the HABs-free equilibrium becomes unstable, as the conditions no longer support the absence of HABs.

Moreover, the proposed model system exhibits bistability behavior within a specific range of the nutrient uptake rate by HABs. This bistability implies that, depending on the initial conditions, the system can converge to one of two distinct stable equilibria. This bistability behavior shows that when the initial density of HABs is low, the model system stabilizes to an equilibrium characterized by a lower density of HABs. Conversely, when the initial density of HABs is high, the system stabilizes to an equilibrium with a higher density of HABs. This behavior highlights the sensitivity of the system to initial conditions and underscores the potential for alternative stable states in aquatic ecosystems influenced by HAB dynamics.

Furthermore, our analysis reveals that when the nutrient uptake rate by HABs exceeds a critical threshold value, the system transitions to a unique stable interior equilibrium. In this regime, the density of HABs increases monotonically with the nutrient uptake rate, reflecting the enhanced ability of HABs to exploit available nutrients and sustain higher population levels. As the uptake rate crosses the threshold value, the system undergoes a Hopf bifurcation, leading to the emergence of stable or unstable limit cycles around the interior equilibrium. This finding illustrates the complex dynamics of the model system, where changes in the nutrient uptake rate can trigger significant shifts in stability behavior, ultimately influencing the long-term ecological outcomes.

While this study provides valuable insights into the dynamics of HABs and their ecological and social impacts, it is important to acknowledge certain limitations. First, the mathematical model simplifies complex real-world interactions by making several assumptions, such as uniform nutrient distribution, homogeneous environmental conditions, and static human population behaviors, which may not fully capture the variability and stochasticity present in natural ecosystems. Additionally, the model does not account for external factors, such as climate change, seasonal variations, or extreme weather events, which can influence HAB dynamics. To address these gaps, future research may integrate more comprehensive datasets, refine model parameters to better capture environmental heterogeneity, and incorporate dynamic socioeconomic factors. Interdisciplinary approaches combining advanced modeling with empirical observations will be crucial for improving predictive accuracy and identifying effective intervention strategies. Ultimately, such efforts will contribute to sustainable management practices that safeguard aquatic ecosystems, protect public health, and support the resilience of communities dependent on these vital resources.

Data Availability The data that support the findings of this study are available within the article.

Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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